Price Externalities and Endogenous Financial Inclusion under Interest-Bearing CBDC

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We study the effects of deposit-like central bank digital currency (CBDC) on equilibrium price dispersion and financial inclusion. When the CBDC rate rises, banked buyers carry more liquidity, inducing sellers to lower prices across the board—a *positive price externality* that benefits unbanked buyers who hold only cash. The same rate increase promotes endogenous financial inclusion, drawing marginal unbanked households into the formal financial system and amplifying the price externality through a composition channel. Endogenous seller adoption further amplifies both effects through a self-reinforcing complementarity between adoption and usage. These findings suggest that the CBDC interest rate shapes the equilibrium price distribution in goods markets and promotes financial inclusion, with benefits extending to agents who do not hold CBDC.

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1. Introduction

This paper studies the effects of a deposit-like central bank digital currency (CBDC) on equilibrium price dispersion and financial inclusion. By altering buyers' liquidity, deposit-like CBDC induces sellers to adjust their posted prices, reshaping the equilibrium price distribution; it also affects households' incentives to participate in the banking system, with implications for financial inclusion. Despite growing policy interest in this design, its implications for goods-market price dispersion and financial access remain largely unexplored in the theoretical literature. We develop a Lagos-Wright monetary model with Burdett-Judd price posting and imperfect substitution between CBDC and bank deposits to fill this gap, and find that deposit-like CBDC generates a positive price externality, promotes financial inclusion, and supports endogenous seller adoption.

We analyze the effects of deposit-like CBDC in an environment that features both equilibrium price dispersion and endogenous financial inclusion. The price-distribution side builds on Head et al. (2012), who introduce a noisy search mechanism—Burdett-Judd price posting—into a Lagos-Wright monetary model: sellers post a single price without observing the buyer's identity, and the equal-profit condition endogenously pins down a non-degenerate equilibrium price distribution. A crucial feature of this mechanism is that all buyers, regardless of their payment portfolio, face the *same* price distribution, because sellers cannot condition their price on the buyer's type. The financial-inclusion side follows Lahcen and Gomis-Porqueras (2021), who model banked and unbanked households as endogenously determined by heterogeneous account-maintenance costs: a buyer participates in the banking system only if the net value of access to financial services exceeds her individual cost. The CBDC side treats CBDC as an imperfect substitute for bank deposits, as in Gong, Han, and Zhu (2023): because CBDC and deposits are accepted in partially distinct sets of meetings, a higher CBDC rate enriches the banked buyer's asset menu rather than merely displacing deposits one-for-one.

Positive price externality and endogenous financial inclusion. Deposit-like CBDC generates a positive price externality through the equal-profit condition. Our model features three payment instruments—cash, bank deposits, and CBDC—with buyers split into banked and unbanked types. The equilibrium price distribution is determined jointly by the liquidity available to each buyer type and the population shares of those types. When the central bank raises the CBDC rate, the holding cost of CBDC falls and banked buyers carry more CBDC balances. Because imperfect substitution preserves a distinct role for deposits, total liquidity rises rather than merely being reallocated. In CBDC-accepting meetings, banked buyers now spend more, raising sellers' expected revenue at each interior posted price. The equal-profit condition requires that this revenue increase be offset by lower prices—shifting the entire price distribution downward in the sense of first-order stochastic dominance.

Since sellers cannot condition their posted price on the buyer's type, the unbanked buyer faces the same lower price distribution despite holding only cash. Her cash purchases more goods on average, raising her consumption and surplus. This spillover—one group's portfolio adjustment improving another group's terms of trade through the price distribution—is the defining feature of the positive price externality under Burdett-Judd price posting, and is absent under bilateral bargaining where the terms of trade depend exclusively on the buyer's own portfolio.

Deposit-like CBDC also promotes financial inclusion. CBDC is available only to banked buyers, so a higher CBDC rate directly raises the net value of maintaining a bank account by enriching the banked buyer's asset menu. Marginal unbanked agents—those whose account-maintenance costs are just above the participation threshold—find it worthwhile to open accounts and gain access to both deposits and CBDC. As the equilibrium banked fraction rises, the component of sellers' expected revenue attributable to CBDC-equipped buyers grows, reinforcing the downward pressure on prices through the equal-profit condition. This composition channel creates a complementarity between financial inclusion and the positive price externality: more banked buyers generate a stronger externality, and a stronger externality raises the incentive to become banked.

Seller adoption of the CBDC payment network is also endogenized. When the CBDC rate is sufficiently high, buyers hold positive CBDC balances regardless of how many sellers have adopted, creating a strictly positive revenue differential for early adopters and breaking the coordination failure that would otherwise trap the economy at zero acceptance. Once adoption begins, wider acceptance raises the return to holding CBDC, inducing buyers to carry larger balances—a self-reinforcing complementarity between adoption and usage that amplifies both the positive price externality and the financial inclusion effect.

These findings have implications for how policymakers should evaluate the design of deposit-like CBDC. The CBDC interest rate is not merely a tool for managing the competitive balance between CBDC and private bank deposits; it can serve as an instrument of financial inclusion. By raising the net value of banking participation, the CBDC rate induces endogenous financial inclusion: marginal unbanked households enter the formal financial system without requiring direct transfers or mandates. The endogenous complementarity between inclusion and the price externality further implies that the policy is self-reinforcing: partial inclusion deepens over time as the composition channel amplifies the initial impulse.

The distributional implications are also broader than a narrow focus on CBDC holders would suggest. Because the price externality operates through sellers' equal-profit condition—which pools revenue across all buyer types—unbanked households benefit from the policy despite holding no CBDC themselves. Evaluating deposit-like CBDC solely through its effects on banked buyers would therefore understate the policy's aggregate impact. We calibrate the model to the U.S. economy and find that these effects are economically significant across all acceptance scenarios, with the largest gains arising when seller adoption is endogenized.

Related literature. This paper contributes to several strands of the literature. Within the New Monetarist CBDC literature, Chiu et al. (2023) study CBDC in a Lagos-Wright model with monopolistic banking and bilateral bargaining, and show that CBDC competition can raise the deposit rate and improve welfare. We build on their banking structure but replace bilateral bargaining with Burdett-Judd price posting, which generates the positive price externality absent under bargaining. Jiang and Zhu (2024) analyze CBDC's pass-through to deposit rates in a competitive banking model; our framework provides a complementary transmission mechanism through the price distribution. Andolfatto (2021) finds that CBDC need not disintermediate banks if the central bank adjusts the CBDC rate appropriately; our model corroborates this result. Keister and Sanches (2022) study optimal CBDC design with heterogeneous agents and show that a properly designed CBDC can improve welfare; our emphasis on price externalities and endogenous participation complements their normative analysis. Gong, Han, and Zhu (2023) study endogenous CBDC acceptance under imperfect substitution between CBDC and deposits; we adopt a similar adoption-cost framework but embed it in a richer environment with price posting. Williamson (2022) develops a model in which CBDC crowds out bank deposits and can lead to disintermediation; our framework shares the concern about deposit displacement but identifies an offsetting benefit through the price externality channel. Davoodalhosseini (2022) characterizes the optimal CBDC design in an economy where CBDC and cash coexist; our analysis adds price posting and banking to this coexistence question.

The paper also relates to the broader macroeconomic literature on CBDC. Brunnermeier and Niepelt (2019) establish an equivalence result showing that CBDC need not alter the allocation if the central bank adjusts its balance sheet appropriately; our model departs from this equivalence because the price-posting friction makes the composition of payment instruments matter for the equilibrium price distribution. Fernández-Villaverde et al. (2021) study CBDC in a Diamond-Dybvig framework and analyze the implications for bank runs and financial stability; our search-theoretic approach complements their analysis by focusing on goods-market outcomes rather than financial fragility. Agur, Ari, and Dell'Ariccia (2022) study optimal CBDC design when agents are heterogeneous in their preferences over payment instruments; our heterogeneity operates through account-maintenance costs and generates an endogenous financial inclusion margin.

Second, the paper contributes to the literature on equilibrium price dispersion in monetary models. Head et al. (2012) introduce Burdett-Judd pricing into a monetary search model and show that money growth affects the level and dispersion of prices; we extend their framework to include CBDC and banking. Wang, Wright, and Liu (2020) study the coexistence of money and credit under price posting; our model adds a third instrument (CBDC) and an explicit banking sector. Burdett, Trejos, and Wright (2017) provide a simplified New Monetarist model with price posting; our contribution to this literature is to show that CBDC policy affects price dispersion through the average buyer composition, a channel distinct from the inflation-cost channel studied in existing work. The pecuniary externality we identify—one agent's portfolio choice affects another's terms of trade through the price distribution—is related to the channel in Kam et al. (2025), who study competitive banking through goods-pricing dispersion.

Third, the paper contributes to the literature on banking, money, and financial inclusion. Berentsen, Camera, and Waller (2007) study money, credit, and banking in a Lagos-Wright framework; Berentsen, Huber, and Marchesiani (2014) extend this to analyze the welfare effects of financial intermediation. Chiu, Dong, and Shao (2018) study the welfare effects of credit arrangements in a search framework. Our model adds CBDC as a third payment instrument alongside cash and bank-intermediated deposits. The financial inclusion extension relates to Lahcen and Gomis-Porqueras (2021), who studies endogenous financial inclusion in a monetary search model and shows that it has important implications for inequality and monetary policy; we show that CBDC amplifies the inclusion margin by enriching the banked buyer's asset menu. Dong and Xiao (2024) study CBDC in an economy with heterogeneous payment needs; our analysis complements theirs by connecting CBDC adoption to equilibrium price dispersion.

Outline. Section 2 presents the model. Section 3 derives the positive price externality and presents calibrated quantitative results. Section 4 endogenizes financial inclusion. Section 5 endogenizes CBDC acceptance. Section 6 concludes. Proofs are in the Appendix.

2. Model

This section presents the model. Section 2.1 describes the environment: agents, preferences, payment instruments, meeting types, and timing. Section 2.2 characterizes the buyer's portfolio problem and derives the Euler equations governing cash, deposit, and CBDC holdings. Section 2.3 derives the seller's pricing decision and the equal-profit condition that pins down the equilibrium price distribution $J(\rho)$. Section 2.4 introduces the banking sector: banks compete à la Cournot in the loan market and take deposits competitively, with CBDC affecting deposit supply and therefore loan volume. Section 2.5 defines stationary monetary equilibrium and establishes existence.

2.1. Environment

Agents and preferences. Time is discrete and continues forever. Each period t is divided into two subperiods. In the first subperiod, agents interact in a frictional decentralized market (DM). In the second subperiod, they interact in a Walrasian centralized market (CM). There are two perishable goods: q in the DM and x in the CM. The common discount factor between consecutive periods is $\beta \in (0, 1)$. We use the notation $X \equiv X_t$ and $X_+ \equiv X_{t+1}$ to denote variables in periods t and $t + 1$, respectively.

There are four types of agents: households, entrepreneurs, N banks, and the government. The household sector consists of a continuum of buyers with measure 1 and a continuum of sellers with measure s. Both buyers and sellers are infinitely lived and participate in the DM and the CM. In the DM, buyers wish to consume but cannot produce, while sellers can produce but do not wish to consume. In the CM, buyers and sellers produce and consume. Entrepreneurs, who are two-period lived, invest when young and consume when old (see below). Agents in the DM are anonymous and lack commitment, precluding credit arrangements.

Financial inclusion. A fraction $\lambda \in (0, 1)$ of buyers are *banked*: they hold a bank account and can carry cash z_B, checkable deposits d_B, and CBDC e_B into the DM. The remaining fraction $1-\lambda$ are *unbanked*: lacking a bank account, they can hold only cash z_U. In Section 4, we endogenize λ by introducing heterogeneous banking costs; until then, λ is treated as exogenous. The distinction between banked and unbanked buyers is central to the price externality mechanism: sellers post a single price before observing the buyer's type, so the equilibrium price distribution reflects the average buyer composition.

A buyer's period utility is

$$\mathcal{U}_b(q, x, h) = u(q) + U(x) - h, \tag{1}$$

where u is the utility from consuming the DM good q, U is the utility from consuming the CM good x, and h denotes labor. We assume $u' > 0$, $u'' < 0$, $u(0) = 0$, and $u'(0) = \infty$. The DM utility takes the CRRA form $u(q) = q^{1-\sigma}/(1 - \sigma)$ with $\sigma \in (0, 1)$, as in Head et al. (2012). The CM utility satisfies $U' > 0$, $U'' < 0$, and $U'(0) = \infty$.

A seller's period utility is

$$\mathcal{U}_s(q, x, h) = -c \cdot q + U(x) - h, \tag{2}$$

where $c > 0$ is the constant marginal cost of producing q in terms of CM goods. All sellers are homogeneous.

Entrepreneurs are two-period lived. They are born in the current CM and die in the next CM. Entrepreneurs cannot work and consume only when old. A young entrepreneur has access to an investment technology that transforms l units of current CM goods into $f(l)$ units of CM goods in the next period, where $f' > 0$, $f'' < 0$, $f'(0) = \infty$, and $f'(\infty) = 0$. Entrepreneurs and households lack mutual commitment and cannot enforce bilateral debt. Banks intermediate between them, as described in Section 2.4.

Payment instruments. There are three instruments that can serve as media of exchange in the DM: fiat money (cash), checkable deposits, and a central bank digital currency (CBDC).[1]

Cash is a physical token issued by the central bank. It pays zero nominal interest. Let z denote a buyer's real cash balances carried into the DM. Checkable deposits are liquid liabilities of banks that can be used as a medium of exchange in the DM. Deposits pay a net real return i_d per period. A buyer entering the DM with real deposit holdings d has $(1 + i_d)d$ units of purchasing power from deposits. The CBDC is a digital liability of the central bank that can also be used for retail payments. It pays a net real return i_e, set as a policy instrument. A buyer entering the DM with real CBDC holdings e has $(1 + i_e)e$ units of purchasing power from the CBDC. Both checkable deposits and the CBDC require electronic payment infrastructure to be usable, but the availability of each may differ across transactions.

Meeting types in the DM. Following Head et al. (2012), we model goods trade as a monetary-exchange version of the noisy search process introduced by Burdett and Judd (1983). Each seller posts a real price ρ and commits to supplying any quantity demanded at that price. Sellers take as given the distribution of posted prices, denoted by $J(\cdot)$, and buyers' demand schedules. Buyers know J but cannot observe individual prices before visiting a seller. Each buyer contacts k sellers with probability α_k. For simplicity, buyers either sample one price with probability $\alpha_1 \in (0, 1)$ or two independent prices with probability $\alpha_2 = 1 - \alpha_1$. If a buyer samples two prices, she purchases from the seller offering the lower price.

All sellers accept all payment instruments. However, the payment instrument that can actually be *used* in a given transaction depends on the type of meeting, which is realized after the match is formed. Following Gong, Han, and Zhu (2023) and Lester, Postlewaite, and Wright (2012), we assume that the usability of each payment instrument in a particular meeting is stochastic. This captures the idea that infrastructure availability, network conditions, and technological constraints vary across transactions, even for the same seller.

Specifically, a banked buyer enters one of four meeting types. With probability Ω_1, only cash is usable (**cash-only meeting**). With probability $\Omega_2(1-\tau)$, only checkable deposits are

[1] We do not consider time deposits. Because CBDC and checkable deposits are imperfect substitutes in our model, there is always positive demand for checkable deposits, and no additional savings instrument is needed to clear the loan market.

TABLE 1. Meeting types and available liquidity (banked buyer)

Meeting type	Probability	Usable instruments	Liquidity ℓ_j
Cash only ($j = m$)	Ω_1	Cash only	z_B
Deposit only ($j = d$)	$\Omega_2(1-\tau)$	Deposit	$(1+i_d)d_B$
Deposit + CBDC ($j = dc$)	$\Omega_2\tau$	Deposit, CBDC	$(1+i_d)d_B + (1+i_e)e_B$
All instruments ($j = a$)	Ω_3	Cash, Dep., CBDC	$z_B + (1+i_d)d_B + (1+i_e)e_B$

The unbanked buyer carries only cash z_U, so her available liquidity is z_U in every meeting type.

usable (**deposit-only meeting**). With probability $\Omega_2\tau$, both deposits and CBDC are usable (**deposit-and-CBDC meeting**). With probability Ω_3, all three instruments are usable (**all-instrument meeting**). The probabilities satisfy $\Omega_1 + \Omega_2 + \Omega_3 = 1$. The parameter $\tau \in [0,1]$ governs the fraction of electronic-payment meetings in which CBDC infrastructure is available. An unbanked buyer carries only cash z_U and therefore has available liquidity z_U in every meeting type.

The probabilities Ω_1, Ω_2, and Ω_3 are calibrated from the Survey of Consumer Payment Choice (SCPC), following the same data source as Gong, Han, and Zhu (2023) and Chiu et al. (2023).

Timing. The timing within each period is as follows.

a. *CM*. Buyers choose portfolios (z_+, d_+, e_+) for the next DM. Entrepreneurs obtain loans from banks and invest. All agents consume x and supply labor h.
b. *DM, stage 1.* Each seller posts a price ρ, not knowing which meeting type will be realized.
c. *DM, stage 2.* Each buyer samples one or two prices from the common distribution $J(\rho)$ and selects the lowest price.
d. *DM, stage 3.* The buyer visits the selected seller. The meeting type is realized: for a banked buyer, cash-only with probability Ω_1, deposit-only with probability $\Omega_2(1-\tau)$, deposit-and-CBDC with probability $\Omega_2\tau$, or all-instrument with probability Ω_3. An unbanked buyer uses cash in all meetings.
e. *DM, stage 4.* The buyer demands quantity q at the posted price ρ, subject to the liquidity constraint determined by the meeting type. Trade occurs.

The key assumption is that the meeting type is realized *after* the seller posts a price but *before* trade occurs. Consequently, at the time of posting, each seller faces the same ex-ante expected payoff regardless of identity, which ensures that the standard Burdett-Judd equal-profit condition applies to a single price distribution $J(\rho)$.

Government. The government is a consolidated monetary and fiscal authority. The central bank issues two forms of liabilities: cash and the CBDC. It stands ready to exchange cash and the CBDC at par in the CM. Let M and $\mathcal{E}$ denote the aggregate nominal stocks of cash and CBDC, respectively. We focus on stationary monetary policies where total nominal liabilities grow at a constant gross rate $\gamma > \beta$, so that $M_+ = \gamma M$ and $\mathcal{E}_+ = \gamma\mathcal{E}$. In a stationary equilibrium the real aggregates $Z \equiv \phi M$ and $E \equiv \phi\mathcal{E}$, where ϕ is the CM price of money, are time-invariant.

The government collects seigniorage revenue from the issuance of new liabilities, pays interest i_e on outstanding CBDC, and finances the residual through lump-sum transfers T to buyers (a negative T represents lump-sum taxes). The government budget constraint is

$$T = (\gamma - 1)(Z + E) - i_e E. \tag{3}$$

2.2. Buyer's Problem

We define the aggregate state as $\mathbf{s} := (Z, E, \gamma, i_e)$ and suppress it from the notation where no confusion arises. This subsection first derives the ex-post DM demand (common to both buyer types), then presents the full CM–DM optimization and Euler equations separately for banked and unbanked buyers.

2.2.1. Ex-post DM Problem

After selecting a seller at price ρ, the meeting type is realized, determining the buyer's available liquidity ℓ. Regardless of buyer type, the CM value function is linear in wealth (shown below for each type), so spending ρq in the DM reduces CM wealth by exactly ρq. The buyer's ex-post problem therefore reduces to maximizing trade surplus:

$$\max_q \{u(q) - \rho q\} \quad \text{s.t.} \quad \rho q \le \ell. \tag{4}$$

Let $\mu \ge 0$ denote the multiplier on the liquidity constraint. The first-order condition is $u'(q) = \rho(1 + \mu)$. When the constraint does not bind ($\mu = 0$), the buyer consumes the unconstrained optimum $q^u(\rho) = \rho^{-1/\sigma}$, where we used $u'(q) = q^{-\sigma}$. When it binds ($\mu > 0$), the buyer spends all available liquidity: $q = \ell/\rho$.

The *cutoff price* $\hat{\rho}(\ell)$ is the price at which unconstrained expenditure exactly equals available liquidity, $\rho \cdot q^u(\rho) = \ell$:

$$\rho^{1-1/\sigma} = \ell \quad \Longrightarrow \quad \hat{\rho}(\ell) = \ell^{\,\sigma/(\sigma-1)}. \tag{5}$$

Since $\sigma \in (0, 1)$, the exponent $\sigma/(\sigma - 1) < 0$, so $\hat{\rho}$ is decreasing in ℓ: more liquidity lowers

TABLE 2. Available liquidity and cutoff prices by meeting type (banked buyer)

Meeting type	Weight w_j	Available liquidity ℓ_j	Cutoff $\hat{\rho}_j$
Cash only ($j = m$)	$\lambda\Omega_1$	z_B	$z_B^{\sigma/(\sigma-1)}$
Deposit only ($j = d$)	$\lambda\Omega_2(1-\tau)$	$(1+i_d)d_B$	$[(1+i_d)d_B]^{\sigma/(\sigma-1)}$
Dep.+CBDC ($j = dc$)	$\lambda\Omega_2\tau$	$(1+i_d)d_B+(1+i_e)e_B$	$[(1+i_d)d_B+(1+i_e)e_B]^{\sigma/(\sigma-1)}$
All instr. ($j = a$)	$\lambda\Omega_3$	$z_B+(1+i_d)d_B+(1+i_e)e_B$	$[z_B+(1+i_d)d_B+(1+i_e)e_B]^{\sigma/(\sigma-1)}$

The unbanked buyer ($j = U$, weight $1-\lambda$) has $\ell_U = z_U$ with cutoff $\hat{\rho}_U = z_U^{\sigma/(\sigma-1)}$.

the cutoff. The optimal consumption is

$$q^*(\rho,\ell) = \begin{cases} \ell/\rho & \text{if } \rho < \hat{\rho}(\ell), \\ \rho^{-1/\sigma} & \text{if } \rho \geq \hat{\rho}(\ell). \end{cases} \tag{6}$$

The binding region corresponds to low prices. This is standard under $\sigma \in (0,1)$: a lower price induces the buyer to demand a larger quantity ($q^u = \rho^{-1/\sigma}$ is decreasing in ρ), which raises desired expenditure $\rho q^u = \rho^{1-1/\sigma}$ and tightens the liquidity constraint.[2]

The meeting types differ only in the available liquidity ℓ. Define the buyer's trade surplus in meeting type j as $S_j(\rho) = u(q_j^*) - \rho q_j^*$, where $q_j^* = q^*(\rho,\ell_j)$. The banked buyer's four meeting types and corresponding liquidity levels are given in Table 1. The unbanked buyer has available liquidity z_U in every meeting.

The cutoff prices depend on the buyer's equilibrium portfolio. Since $\hat{\rho}$ is decreasing in ℓ, the ranking of cutoffs across meeting types is determined by the relative magnitudes of the portfolio components. For instance, when $e_B > 0$ and $z_B > 0$, we have $\ell_d < \ell_{dc} < \ell_a$ and hence $\hat{\rho}_d > \hat{\rho}_{dc} > \hat{\rho}_a$; but when $e_B = 0$, the deposit-only and deposit-and-CBDC meetings yield identical liquidity $\ell_d = \ell_{dc}$ and identical cutoffs.

Let $J(\rho)$ denote the distribution of posted prices (derived in Section 2.3). Since the buyer samples one price with probability α_1 and two prices with probability $\alpha_2 = 1-\alpha_1$, the distribution of the *transaction price* (the lowest sampled price) is

$$G(\rho) = \alpha_1 J(\rho) + \alpha_2\left[1 - (1 - J(\rho))^2\right]. \tag{7}$$

The price distribution $G(\rho)$ is the same for banked and unbanked buyers. The meeting type affects only the liquidity constraint—i.e., ℓ_j—not which prices the buyer faces. This is the key simplification from making sellers homogeneous and the meeting type stochastic, and it restores full consistency with the Burdett-Judd framework.

[2]See Head et al. (2012) for a detailed discussion of this property.

2.2.2. Banked Buyer

The banked buyer (fraction λ of the population) has access to three payment instruments: cash z_B, checkable deposits d_B, and CBDC e_B. We solve the banked buyer's problem backward, starting from the CM.

CM problem. Let W^B and V^{bk} denote the banked buyer's value functions in the CM and DM, respectively. The banked buyer enters the CM holding a portfolio (z, d, e) of real cash balances, real checkable deposits, and real CBDC balances. In the CM, the buyer chooses consumption x, labor h, and a portfolio (z_+, d_+, e_+) to carry into the next DM:

$$W^B(z, d, e) = \max_{x,h,z_+,d_+,e_+} \left\{U(x) - h + \beta V^{bk}(z_+, d_+, e_+)\right\} \tag{8}$$

subject to

$$x + \gamma z_+ + \gamma d_+ + \gamma e_+ = T + h + z + (1 + i_d)d + (1 + i_e)e + \Pi, \tag{9}$$

where $\Pi \geq 0$ denotes bank dividends received by the buyer (specified in Section 2.4). On the right-hand side, T is the government transfer, h is labor income at unit wage, the remaining terms are the real value of assets brought from the previous DM and bank dividends. On the left-hand side, acquiring real balances z_+ for the next period costs γz_+ in current CM goods, because the nominal price level grows at gross rate γ between periods; the same applies to d_+ and e_+.

Substituting the budget constraint for h, the problem separates:

$$\begin{aligned} W^B(z, d, e) = \max_x \{U(x) - x\} + \max_{z_+,d_+,e_+} &\left\{-\gamma z_+ - \gamma d_+ - \gamma e_+ + \beta V^{bk}(z_+, d_+, e_+)\right\} \\ &+ T + z + (1 + i_d)d + (1 + i_e)e + \Pi. \end{aligned} \tag{10}$$

Two properties follow immediately. First, CM consumption satisfies

$$U'(x^*) = 1, \tag{11}$$

independent of the portfolio. Second, the portfolio choice (z_+, d_+, e_+) is independent of current wealth, so all banked buyers select the same portfolio—the standard degeneracy result from quasi-linear preferences (Lagos and Wright 2005). The first-order conditions for the portfolio are

$$\beta V_z^{bk}(z_+, d_+, e_+) = \gamma, \tag{12}$$

$$\beta V_d^{bk}(z_+, d_+, e_+) \leq \gamma, \quad \text{with equality if } d_+ > 0, \tag{13}$$

$$\beta V_e^{bk}(z_+, d_+, e_+) \le \gamma, \quad \text{with equality if } e_+ > 0. \tag{14}$$

The cash condition (12) holds as a strict equality because cash always carries a positive liquidity premium under $\gamma > \beta$. Conditions (13) and (14) are complementary slackness conditions: the buyer holds the asset if and only if the discounted marginal value of liquidity justifies the cost γ. The envelope conditions are

$$W_z^B = 1, \qquad W_d^B = (1 + i_d), \qquad W_e^B = (1 + i_e). \tag{15}$$

DM value function. Conditional on the transaction price ρ, the meeting type is realized independently. The banked buyer's ex-ante DM value function is

$$V^{bk}(z_B, d_B, e_B) = \int_{\underline{\rho}}^{\bar{\rho}} [\Omega_1 S_m(\rho) + \Omega_2(1-\tau) S_d(\rho) + \Omega_2 \tau S_{dc}(\rho) + \Omega_3 S_a(\rho)] \, dG(\rho) + W^B(z_B, d_B, e_B), \tag{16}$$

where $S_j(\rho) = u(q_j^*) - \rho q_j^*$ is the trade surplus in meeting type j and $[\underline{\rho}, \bar{\rho}]$ is the support of J. The four surplus terms correspond to cash-only, deposit-only, deposit-and-CBDC, and all-instrument meetings, weighted by the respective meeting probabilities.

Euler equations. Taking derivatives of (16) with respect to z_B, d_B, and e_B, and combining with the CM first-order conditions (12)–(14) and the envelope conditions (15), we obtain the banked buyer's Euler equations. Define the marginal cost of holding each asset:

$$\phi_z \equiv \frac{\gamma - \beta}{\beta}, \tag{17}$$

$$\phi_d \equiv \frac{\gamma - (1 + i_d)\beta}{(1 + i_d)\beta}, \tag{18}$$

$$\phi_e \equiv \frac{\gamma - (1 + i_e)\beta}{(1 + i_e)\beta}. \tag{19}$$

These represent the net cost of carrying one unit of real purchasing power in each instrument into the DM. Cash bears the full inflation cost $\phi_z = (\gamma - \beta)/\beta$; the interest returns on deposits and the CBDC partially offset this cost. Define the marginal liquidity benefit $\Lambda(\ell, G) \equiv \int_{\underline{\rho}}^{\hat{\rho}(\ell)} (u'(q^*)/\rho - 1) \, dG(\rho)$. The three Euler equations are:

Cash. Cash provides liquidity in cash-only meetings (weight Ω_1) and all-instrument meetings (weight Ω_3):

$$\phi_z = \Omega_1 \Lambda(z_B, G) + \Omega_3 \Lambda(\ell_a, G). \tag{20}$$

Deposits. Deposits provide liquidity in deposit-only, deposit-and-CBDC, and all-instrument meetings:

$$\phi_d = \Omega_2(1-\tau)\Lambda(\ell_d, G) + \Omega_2\tau\Lambda(\ell_{dc}, G) + \Omega_3\Lambda(\ell_a, G). \tag{21}$$

CBDC. The CBDC provides liquidity in deposit-and-CBDC and all-instrument meetings:

$$\phi_e \leq \Omega_2\tau\Lambda(\ell_{dc}, G) + \Omega_3\Lambda(\ell_a, G), \tag{22}$$

with equality if $e_B > 0$. The CBDC is not usable in deposit-only or cash-only meetings, so its marginal benefit is lower than that of deposits, ceteris paribus.

2.2.3. Unbanked Buyer

The unbanked buyer (fraction $1-\lambda$ of the population) lacks a bank account and can hold only cash z_U. This buyer has no access to deposits or CBDC.

CM problem. Let W^U and V^U denote the unbanked buyer's value functions in the CM and DM, respectively. The unbanked buyer enters the CM holding cash z and chooses consumption x, labor h, and cash $z_{U,+}$ for the next DM:

$$W^U(z) = \max_{x,h,z_{U,+}} \left\{ U(x) - h + \beta V^U(z_{U,+}) \right\} \tag{23}$$

subject to $x + \gamma z_{U,+} = T + h + z + \Pi$. Substituting for h, the problem separates into CM consumption ($U'(x^*) = 1$) and a portfolio problem in $z_{U,+}$ alone. Quasi-linearity implies that all unbanked buyers choose the same z_U, and the envelope condition is $W^U_z = 1$.

DM value function. The unbanked buyer faces the same transaction price distribution $G(\rho)$ as the banked buyer, but holds only cash z_U as liquidity in every meeting type. Her ex-ante DM value function is

$$V^U(z_U) = \int_{\underline{\rho}}^{\bar{\rho}} S_U(\rho)\, \mathrm{d}G(\rho) + W^U(z_U), \tag{24}$$

where $S_U(\rho) = u(q^*_U) - \rho q^*_U$ with $q^*_U = q^*(\rho, z_U)$. Since the unbanked buyer cannot hold deposits or CBDC, the meeting type is irrelevant for her: available liquidity is z_U regardless of which payment infrastructure is present.

Euler equation. The unbanked buyer's first-order condition $\beta V^U_z = \gamma$, combined with the envelope condition $W^U_z = 1$, yields:

$$\phi_z = \Lambda(z_U, G). \tag{25}$$

Since $\gamma > \beta$, we have $\phi_z > 0$, and the Inada condition $u'(0) = \infty$ ensures $\Lambda(0, G) = \infty$, so $z_U > 0$ in equilibrium. Comparing (25) with the banked buyer's cash Euler (20), the unbanked buyer equates the marginal cost ϕ_z to the liquidity benefit from cash in a single meeting type, whereas the banked buyer's marginal liquidity benefit from cash is the sum of two terms—from cash-only meetings (weight Ω_1) and all-instrument meetings (weight Ω_3)—because cash provides purchasing power in both meeting types. The marginal cost ϕ_z is identical for both buyer types; the difference lies in the structure of the marginal benefit. In general, $z_U \neq z_B$: the two buyer types hold different cash balances even though they face the same transaction prices.

2.2.4. Portfolio Characterization

The Euler equations (20)–(22) and (25) jointly determine the equilibrium portfolios (z_B, d_B, e_B) and z_U. The following lemma characterizes how the banked buyer's portfolio composition depends on i_e.

LEMMA 1. *Suppose $\gamma > \beta$, $\tau \in (0, 1)$, and $i_e < \bar{i}_e \equiv \gamma/\beta - 1$. There exists a threshold $\underline{i}_e$ such that:*

(i) *If $i_e \leq \underline{i}_e$, the banked buyer holds $e_B = 0$, $d_B > 0$, and $z_B > 0$.*

(ii) *If $i_e > \underline{i}_e$, the banked buyer holds $e_B > 0$, $d_B > 0$, and $z_B > 0$. Both* (21) *and* (22) *hold as equalities.*

In both cases, the cash Euler equation (20) *holds as an equality, and the unbanked buyer holds $z_U > 0$ satisfying* (25). *The threshold $\underline{i}_e$ is the unique value satisfying $\phi_e(\underline{i}_e) = \Omega_2 \tau \Lambda(\ell_d^0, G) + \Omega_3 \Lambda(\ell_a^0, G)$, where (ℓ_d^0, ℓ_a^0) are the equilibrium liquidity levels at $e_B = 0$. The threshold satisfies $\underline{i}_e < \bar{i}_e$ and may be negative: if $\phi_e(0) < \Omega_2 \tau \Lambda(\ell_d^0, G) + \Omega_3 \Lambda(\ell_a^0, G)$, then $e_B > 0$ for all $i_e \geq 0$ and case (i) never applies.*

When i_e is low, the marginal cost of holding CBDC exceeds its liquidity benefit, so banked buyers rely entirely on deposits for electronic payments. As i_e rises past $\underline{i}_e$, the holding cost of CBDC falls sufficiently to justify positive holdings, and both deposits and the CBDC coexist. The key feature is that deposits are never fully crowded out: the Inada condition $u'(0) = \infty$ ensures that deposits have infinite marginal value in deposit-only meetings (which occur with probability $\Omega_2(1 - \tau) > 0$ when $\tau < 1$). Hence imperfect substitutability—driven by $\tau < 1$—sustains a strictly positive demand for deposits at any CBDC interest rate. Full crowding out ($d_B = 0$) is possible only in the knife-edge case $\tau = 1$, where deposit-only meetings vanish and deposits and the CBDC become perfect substitutes; throughout this paper we maintain $\tau < 1$. The bound $\bar{i}_e = \gamma/\beta - 1$ ensures $\phi_e \geq 0$; if $i_e \geq \bar{i}_e$, the holding cost of CBDC becomes non-positive and demand is unbounded, so we maintain $i_e < \bar{i}_e$ throughout.

Proof. See Appendix A.1. □

2.3. Seller's Problem

We solve the seller's problem backward, starting from the CM.

2.3.1. Seller's Optimization in the CM

The seller's CM value function is

$$W^S(z, d, e) = \max_{x,h} \left\{ U(x) - h + \beta V^S \right\} \tag{26}$$

subject to $x = h + z + (1 + i_d)d + (1 + i_e)e$, where (z, d, e) are the payment balances received in the previous DM. As with the buyer, the seller's optimal CM consumption satisfies $U'(x^*) = 1$, and W^S is linear in wealth:

$$W^S_z = 1, \qquad W^S_d = (1 + i_d), \qquad W^S_e = (1 + i_e). \tag{27}$$

Since sellers do not consume in the DM, carrying assets into the next period would incur the holding cost γ per unit of purchasing power without generating any liquidity benefit; hence sellers optimally set $z_+ = d_+ = e_+ = 0$. The linearity of W^S also implies that sellers are indifferent over which payment instrument they receive in the DM—one unit of real purchasing power in any instrument converts to one unit of CM goods. This is why the seller's pricing decision does not depend on the composition of the buyer's payment.

2.3.2. Seller's Optimization in the DM

A seller posting price ρ earns profit $R_j(\rho) = q^*_j(\rho, \ell_j) \cdot (\rho - c)$ in meeting type j, where q^*_j is the buyer's optimal demand derived in Section 2.2.1, ℓ_j is the buyer's available liquidity in meeting type j (Table 2), and c is the constant marginal cost of DM production. Substituting the demand schedule (6):

$$R_j(\rho) = \begin{cases} \ell_j\left(1 - \dfrac{c}{\rho}\right) & \text{if } \rho < \hat{\rho}_j, \\ \rho^{-1/\sigma}(\rho - c) & \text{if } \rho \geq \hat{\rho}_j. \end{cases} \tag{28}$$

In the constrained region ($\rho < \hat{\rho}_j$), the buyer spends all available liquidity ℓ_j regardless of the price, so R_j is strictly increasing in ρ. In the unconstrained region ($\rho \geq \hat{\rho}_j$), the buyer freely optimizes at $q^u(\rho) = \rho^{-1/\sigma}$, and the intensive margin $R_j(\rho) = \rho^{-1/\sigma}(\rho - c)$ is single-peaked (quasi-concave) with a unique maximum at $\rho^m \equiv c/(1 - \sigma)$.

At the time of posting, the seller does not observe which buyer type or meeting type will be realized. Since all sellers are homogeneous and the meeting type is independent of seller identity (Section 2.1), every seller faces the same *expected intensive margin*, which aggregates over all buyer states:

$$\bar{R}(\rho) = w_U\, R_U(\rho) + w_m\, R_m(\rho) + w_d\, R_d(\rho) + w_{dc}\, R_{dc}(\rho) + w_a\, R_a(\rho), \tag{29}$$

where $R_j(\rho)$ is evaluated at liquidity ℓ_j from Table 1, with $R_U(\rho)$ evaluated at the unbanked buyer's cash z_U, and

$$w_U \equiv 1 - \lambda, \quad w_m \equiv \lambda\Omega_1, \quad w_d \equiv \lambda\Omega_2(1-\tau), \quad w_{dc} \equiv \lambda\Omega_2\tau, \quad w_a \equiv \lambda\Omega_3. \tag{30}$$

The five weights sum to unity. The unbanked buyer's weight $w_U = 1 - \lambda$ ensures that the seller's pricing decision reflects the financial composition of the buyer population. A seller posting price ρ trades with probability that depends on its rank in the price distribution. Under the Burdett-Judd search protocol, with probability α_1 the buyer sampled only one price and trades for certain; with probability α_2, the buyer sampled two prices and trades with this seller only if ρ is the lower, which occurs with probability $2(1 - J(\rho))$ since J is atomless. The seller's expected profit is therefore

$$\Pi(\rho) = \underbrace{\left[\alpha_1 + 2\alpha_2(1 - J(\rho))\right]}_{\text{extensive margin}} \cdot \underbrace{\bar{R}(\rho)}_{\text{expected intensive margin}}. \tag{31}$$

The seller posting the highest price $\bar{\rho}$ in the support of J has $J(\bar{\rho}) = 1$ and trades only with single-sampling buyers, earning $\Pi^m(\bar{\rho}) = \alpha_1 \cdot \bar{R}(\bar{\rho})$. Since the extensive margin is fixed at α_1, this seller maximizes the expected intensive margin, so the monopoly price is $\bar{\rho} = \arg\max_\rho \bar{R}(\rho)$. The following proposition characterizes $\bar{\rho}$.

PROPOSITION 1. *The expected intensive margin $\bar{R}(\rho)$ has a unique maximizer $\bar{\rho} \in (c, \infty)$.*

(i) *$\bar{R}$ is strictly increasing on $(c, \bar{\rho})$ and strictly decreasing on $(\bar{\rho}, \infty)$.*

(ii) *$\bar{\rho} \geq \rho^m \equiv c/(1-\sigma)$, with equality if and only if $\hat{\rho}_j \leq \rho^m$ for all j (i.e., no meeting type is liquidity-constrained at ρ^m).*

(iii) *When $\bar{\rho} > \rho^m$, let $\mathcal{C} \equiv \{j : \hat{\rho}_j > \bar{\rho}\}$ and $\mathcal{U} \equiv \{j : \hat{\rho}_j \leq \bar{\rho}\}$ denote the constrained and unconstrained meeting types at $\bar{\rho}$. Then $\bar{\rho}$ is the unique solution to*

$$\frac{\xi c}{\bar{\rho}^2} + R'_{unc}(\bar{\rho}) = 0, \tag{32}$$

where $R_{unc}(\rho) \equiv \rho^{-1/\sigma}(\rho - c)$ and

$$\xi \equiv \frac{\sum_{j \in \mathcal{C}} w_j\, \ell_j}{\sum_{j \in \mathcal{U}} w_j}. \tag{33}$$

Moreover, for a fixed partition $(\mathcal{C}, \mathcal{U})$, $\bar{\rho}$ is strictly increasing in ξ.

The key force behind part (ii) is the *cost-reduction effect* from constrained buyers. At ρ^m, the unconstrained intensive margin is at its peak, so $R'_{unc}(\rho^m) = 0$. But each constrained meeting type contributes $R'_j(\rho^m) = \ell_j c/(\rho^m)^2 > 0$ to $\bar{R}'(\rho^m)$, because a constrained buyer's expenditure is fixed at ℓ_j and raising the price only lowers the seller's production cost. Hence $\bar{R}'(\rho^m) > 0$ whenever at least one meeting type is constrained at ρ^m, and the maximum lies strictly above ρ^m. At $\bar{\rho}$, this cost-reduction force (the term $\xi c/\bar{\rho}^2 > 0$ in the first-order condition) is exactly offset by the demand-reduction force from unconstrained buyers ($R'_{unc}(\bar{\rho}) < 0$).

The parameter ξ is a sufficient statistic for the influence of liquidity constraints on the monopoly price. It aggregates the weighted liquidity of constrained meeting types, normalized by the total weight of unconstrained types. When $\xi = 0$ (no constrained meetings), $\bar{\rho} = \rho^m$ as in the standard Burdett-Judd model. When $\xi > 0$, the monopoly price exceeds ρ^m, and is strictly increasing in ξ: a larger share of constrained meetings amplifies the cost-reduction effect, allowing sellers to sustain higher prices. Importantly, ξ depends on the buyer's equilibrium portfolio through the liquidities of constrained meeting types. Under (MA), the deposit-and-CBDC and all-instrument types are unconstrained at $\bar{\rho}$, so ξ aggregates the liquidities of the remaining constrained types (cash-only and, depending on parameters, deposit-only). The CBDC interest rate i_e affects the monopoly price indirectly by altering equilibrium portfolios and potentially shifting which meeting types are constrained at $\bar{\rho}$. The equilibrium determination of these objects is analyzed in Section 2.5.

Proof. See Appendix A.5. □

Maintained assumption (MA). Throughout the analysis we assume that $\hat{\rho}_{dc} \leq \bar{\rho}$ in every equilibrium considered, where $\hat{\rho}_{dc}$ is the constrained-unconstrained cutoff for the deposit-and-CBDC buyer type. This condition ensures that *dc*- and *a*-type buyers are unconstrained at the monopoly price, which is the configuration underlying both the FOSD result (Proposition 3, Step 2) and the comparative-static analysis of Section 3. Since $\hat{\rho}_a < \hat{\rho}_{dc}$ whenever $\ell_{dc} < \ell_a$ (recall that $\hat{\rho}$ is decreasing in ℓ), the condition $\hat{\rho}_{dc} \leq \bar{\rho}$ implies $\hat{\rho}_a \leq \hat{\rho}_{dc} \leq \bar{\rho}$ as well. The assumption is verified numerically under the baseline calibration for all $i_e \in [0, \bar{i}_e)$.

In equilibrium, all prices in the support $[\underline{\rho}, \bar{\rho}]$ of J must yield the same expected profit; otherwise, a seller would deviate to a higher-profit price. The equal-profit condition $\Pi(\rho) = \Pi(\bar{\rho})$ for all $\rho \in [\underline{\rho}, \bar{\rho}]$ yields

$$[\alpha_1 + 2\alpha_2(1 - J(\rho))]\, \bar{R}(\rho) = \alpha_1 \bar{R}(\bar{\rho}). \tag{34}$$

LEMMA 2. *Given search frictions $\alpha_1, \alpha_2 \in (0,1)$ and the equilibrium portfolios (z_B, d_B, e_B)*

and z_U, the posted-price distribution consistent with profit maximization by all sellers is

$$J(\rho) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\frac{\bar{R}(\bar{\rho})}{\bar{R}(\rho)} - 1\right], \qquad \rho \in [\underline{\rho}, \bar{\rho}], \tag{35}$$

where the upper bound $\bar{\rho} = \arg\max \bar{R}(\rho)$ is the monopoly price characterized in Proposition 1, and the lower bound $\underline{\rho}$ is determined by $J(\underline{\rho}) = 0$:

$$\bar{R}(\underline{\rho}) = \frac{\alpha_1}{\alpha_1 + 2\alpha_2}\bar{R}(\bar{\rho}). \tag{36}$$

The distribution J is continuous and strictly increasing on $[\underline{\rho}, \bar{\rho}]$.

Proof. See Appendix A.2. □

The equal-profit condition yields the standard Burdett-Judd price dispersion. A seller posting $\bar{\rho}$ trades only with single-sampling buyers (low volume, high margin). A seller posting $\underline{\rho}$ trades with all visiting buyers (high volume, low margin). Every price in $[\underline{\rho}, \bar{\rho}]$ yields the same expected profit, so sellers are indifferent across the support. Crucially, because all sellers are homogeneous and face the same expected intensive margin $\bar{R}(\rho)$, there is a single price distribution $J(\rho)$ rather than type-specific distributions. This single-distribution structure is fully consistent with the Burdett-Judd framework.

2.4. Bank's Problem

2.4.1. Loan Demand from Entrepreneurs

Entrepreneurs are two-period lived: born in the current CM and dying in the next CM. A young entrepreneur has no endowment but has access to an investment technology f that transforms l units of current CM goods into $f(l)$ units of next-period CM goods, where $f' > 0$, $f'' < 0$, $f'(0) = \infty$, and $f'(\infty) = 0$. Since entrepreneurs lack endowments, they borrow from banks at the gross loan rate $R_l \equiv 1 + i_l$. The entrepreneur's problem is

$$\max_{l \geq 0} \{f(l) - R_l l\}. \tag{37}$$

The first-order condition $f'(l) = R_l$ defines the *aggregate loan demand*

$$L^d(R_l) \equiv f'^{-1}(R_l), \tag{38}$$

which is strictly decreasing in R_l. Equivalently, the *inverse loan demand* is $R_l = f'(L)$.

ASSUMPTION 1. $2f''(L) + f'''(L)L < 0$ *for all* $L > 0$.

Assumption 1 ensures uniqueness of the symmetric Cournot equilibrium below. It requires the slope of each bank's marginal revenue to be negative, which makes the best-response mapping a contraction (cf. Gong, Han, and Zhu 2023). The functional form $f(l) = Al^{\eta}$ with $\eta \in (0,1)$ satisfies Assumption 1.

2.4.2. Cournot Competition in the Loan Market

There are $N \geq 2$ identical banks, owned by households. Each bank j accepts checkable deposits d_j from households at the competitive gross deposit rate $R_d \equiv 1 + i_d$ and must hold a fraction $\chi \in (0,1)$ of deposits as reserves at the central bank, earning gross reserve rate $R_r \equiv 1 + i_r$. The remaining fraction $(1-\chi)$ is extended as loans to entrepreneurs: $l_j = (1-\chi)d_j$. Bank j's per-period profit is

$$\Pi_j = R_l(L_{-j} + l_j) \cdot l_j + R_r \cdot \chi\, d_j - R_d \cdot d_j - \eta_d \cdot d_j, \tag{39}$$

where $\eta_d \geq 0$ is a per-unit deposit handling cost and $L_{-j} \equiv \sum_{k \neq j} l_k$.

The deposit market is competitive: each bank takes R_d as given. In the loan market, banks compete in quantities à la Cournot. Since $l_j = (1-\chi)d_j$, choosing l_j is equivalent to choosing d_j. The first-order condition for l_j is

$$R_l(L_{-j} + l_j) + R_l'(L_{-j} + l_j) \cdot l_j = \frac{R_d + \eta_d - \chi R_r}{1-\chi}. \tag{40}$$

The left-hand side is bank j's marginal revenue from lending. The right-hand side is the effective marginal funding cost: a unit of loans requires $1/(1-\chi)$ units of deposits, each costing $R_d + \eta_d$, net of reserve interest $\chi R_r/(1-\chi)$. Under Assumption 1, the second-order condition $2R_l' + R_l'' \cdot l_j < 0$ holds and the Cournot equilibrium exists and is unique.

Symmetric equilibrium. In the symmetric equilibrium, $l_j = L/N$ for all j. Substituting into (40) and using $R_l(L) = f'(L)$:

$$f'(L) + f''(L) \cdot \frac{L}{N} = \frac{R_d + \eta_d - \chi R_r}{1-\chi}. \tag{41}$$

Rearranging, the equilibrium deposit rate satisfies

$$i_d = (1-\chi)\left[i_l + f''(L) \cdot \frac{L}{N}\right] + \chi\, i_r - \eta_d. \tag{42}$$

Since $f'' < 0$, the spread between the loan rate and the deposit rate is

$$i_l - i_d = \chi(i_l - i_r) + \eta_d - (1-\chi)f''(L) \cdot \frac{L}{N} > 0. \tag{43}$$

The spread has three components: the reserve tax $\chi(i_l - i_r)$ (foregone lending on required reserves), the handling cost η_d, and the Cournot markup $-(1-\chi)f''(L) \cdot L/N > 0$. As $N \to \infty$, the Cournot markup vanishes and (42) reduces to $i_d = (1-\chi)i_l + \chi i_r - \eta_d$, recovering the competitive benchmark as a special case.

Bank profits and deposit market clearing. At the symmetric Cournot equilibrium, each bank earns

$$\Pi_j = \left[(1-\chi)i_l + \chi i_r - i_d - \eta_d\right] \cdot \frac{D}{N} = -f''(L) \cdot \left(\frac{L}{N}\right)^2 > 0. \tag{44}$$

Total bank profits $\Pi = N \cdot \Pi_j = -f''(L) \cdot L^2/N$ are distributed as dividends to households and appear in the buyer's CM budget constraint (9). Aggregate loans equal the loanable portion of aggregate deposits:

$$L = (1-\chi)D, \tag{45}$$

where $D \equiv \lambda\, d_B$ denotes aggregate checkable deposits. The deposit rate i_d^* is determined in equilibrium by the banked buyer's deposit Euler equation (21); the Cournot condition (42) then determines the loan rate i_l^*.

2.5. Stationary Monetary Equilibrium

We focus on stationary equilibria in which all real variables are constant over time and all nominal variables grow at the common gross rate γ. In such an equilibrium, the buyer's portfolio (z, d, e), the price distribution $J(\rho)$, and all interest rates are time-invariant.

DEFINITION 1. *Given monetary policy (γ, i_e) with $\gamma > \beta$ and $i_e < \bar{i}_e \equiv \gamma/\beta - 1$, the banked fraction $\lambda \in (0,1)$, the CBDC acceptance rate $\tau \in (0,1)$, and the number of banks $N \geq 2$, a* ***stationary monetary equilibrium*** *is a tuple*

$$\left(z_B^*,\, d_B^*,\, e_B^*,\, z_U^*,\, L^*,\, J^*(\rho),\, i_d^*,\, i_l^*\right)$$

such that the following conditions are satisfied.

(i) ***Buyer optimization.*** *The banked buyer's portfolio (z_B^*, d_B^*, e_B^*) satisfies the Euler equations* (20)–(22) *given (J^*, i_d^*, i_e), and the unbanked buyer's cash z_U^* satisfies* (25)*. The optimal DM demand in each meeting type is $q_j^*(\rho) = q^*(\rho, \ell_j^*)$ from equation* (6)*, with liquidity levels*

$$\ell_m^* = z_B^*, \quad \ell_d^* = (1+i_d^*)d_B^*, \quad \ell_{dc}^* = (1+i_d^*)d_B^* + (1+i_e)e_B^*, \quad \ell_a^* = z_B^* + \ell_{dc}^*, \quad \ell_U^* = z_U^*,$$

for $j \in \{U, m, d, dc, a\}$, and cutoffs $\hat{\rho}_j^* = (\ell_j^*)^{\sigma/(\sigma-1)}$. CM consumption satisfies $U'(x^*) = 1$.

(ii) ***Seller optimization.*** *The price distribution $J^*(\rho)$ satisfies the equal profit condition* (34) *given the equilibrium portfolios. Equivalently,* J^* *is given by Lemma 2:*

$$J^*(\rho) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\frac{\bar{R}(\bar{\rho}^*)}{\bar{R}(\rho)} - 1\right], \qquad \rho \in [\underline{\rho}^*, \bar{\rho}^*],$$

where $\bar{R}$ *is the five-term expected intensive margin* (29) *and* $\bar{\rho}^* = \arg\max \bar{R}(\rho)$ *is the monopoly price (Proposition 1).*

(iii) ***Entrepreneur optimization.*** *Total loans* L^* *and the loan rate* i_l^* *satisfy* $f'(L^*) = 1 + i_l^*$.

(iv) ***Cournot equilibrium in the loan market.*** *The deposit rate and loan rate satisfy the symmetric Cournot condition* (42)*:*

$$i_d^* = (1-\chi)\left[i_l^* + f''(L^*)\cdot\frac{L^*}{N}\right] + \chi\, i_r - \eta_d. \tag{46}$$

(v) ***Deposit market clearing.*** *Aggregate loans equal the loanable portion of aggregate deposits:*

$$(1-\chi)\lambda\, d_B^* = L^*. \tag{47}$$

(vi) ***Government budget balance.*** *The lump-sum transfer satisfies* (3)*:* $T = (\gamma - 1)(Z + E) - i_e E$.

PROPOSITION 2 (Existence). *Under the maintained assumptions* ($\gamma > \beta$, $\tau \in (0,1)$, $i_e < \bar{i}_e$, $N \geq 2$, *Assumption 1), a stationary monetary equilibrium exists.*

Proof. See Appendix A.10. The proof reduces the system to a fixed-point in deposits d_B. Given d_B, the Cournot condition pins down i_d, the Euler equations and equal-profit condition determine the portfolio and price distribution as a fixed point on the space of CDFs (by Schauder's theorem), and the deposit Euler equation yields an excess-cost function that changes sign, so the intermediate value theorem delivers a root. □

3. Positive Price Externality of CBDC

This section establishes the central mechanism of the paper. Section 3.1 shows that raising the CBDC rate compresses the equilibrium price distribution through a first-order stochastic dominance shift, benefiting all buyers including the unbanked. Section 3.2 calibrates the model to the pre-CBDC U.S. economy and evaluates the quantitative magnitude of the externality.

3.1. Positive Price Externality

Consider an increase in the CBDC rate i_e above the entry threshold $\underline{i}_e$ (Lemma 1), so that banked buyers begin holding positive CBDC balances. This subsection traces the resulting chain of effects: the portfolio adjustment raises total buyer liquidity in CBDC-accepting meetings, the higher liquidity shifts the seller's expected intensive margin $\bar{R}(\rho)$

upward at interior prices, and the equal-profit condition forces the equilibrium price distribution to place more weight at lower prices. Because sellers cannot observe the buyer's type or available liquidity when posting a price, the downward price shift benefits all buyers—including the unbanked, who hold no CBDC.

When the central bank raises i_e, the opportunity cost of holding CBDC falls: each unit earns more interest, narrowing the gap between its return and the rate of time preference. Facing a lower cost, the banked buyer carries more e_B into the DM. Unlike a reallocation between existing instruments, this increase represents the activation of a new payment channel. The buyer now enters some DM meetings with purchasing power that was previously unavailable—not merely shifted from deposits or cash.

Deposits do decline as banked buyers partially substitute toward CBDC, but deposits retain an exclusive role: in the $(1-\tau)$-fraction of electronic meetings where the CBDC network is unavailable, deposits are the only electronic means of payment. Subtracting the CBDC Euler equation (22) from the deposit Euler equation (21) confirms this coexistence:

$$\phi_d - \phi_e \;=\; \Omega_2(1-\tau)\,\Lambda\big((1+i_d)\,d_B,\; G\big). \tag{48}$$

The cost premium of deposits over CBDC ($\phi_d - \phi_e > 0$) is exactly justified by the marginal liquidity benefit that deposits provide in deposit-only meetings. As long as these meetings occur with positive probability ($\tau < 1$), the buyer optimally holds both instruments. When $\tau = 1$, the CBDC is accepted everywhere deposits are, and the two instruments become perfect substitutes; the buyer holds whichever offers the higher return.

The net effect on total buyer liquidity is positive.

LEMMA 3. *In any equilibrium with $e_B > 0$, a marginal increase in i_e raises the banked buyer's available liquidity in every meeting type where the CBDC is accepted:*

$$\frac{\partial}{\partial i_e}\,\ell_{dc} = \frac{\partial}{\partial i_e}\big[(1+i_d)\,d_B + (1+i_e)\,e_B\big] > 0, \tag{49}$$

$$\frac{\partial}{\partial i_e}\,\ell_a = \frac{\partial}{\partial i_e}\big[z_B + (1+i_d)\,d_B + (1+i_e)\,e_B\big] > 0. \tag{50}$$

The banked buyer's cash holdings decline ($\partial z_B/\partial i_e < 0$), but the increase in electronic liquidity (49) *more than compensates, so that all-instrument liquidity ℓ_a also rises.*

Proof. See Appendix A.3. □

The lemma captures the central mechanism: CBDC does not merely crowd out deposits or cash one-for-one. The CBDC Euler equation (22) requires the weighted marginal liquidity benefit $\Omega_2\tau\,\Lambda(\ell_{dc}, G) + \Omega_3\,\Lambda(\ell_a, G)$ to equal ϕ_e, which is decreasing in i_e. A lower ϕ_e requires a lower marginal benefit, which in turn requires higher liquidity in CBDC-accepting meetings. The gain in $(1+i_e)\,e_B$ more than offsets any decline in $(1+i_d)\,d_B$, establishing (49).

The cash Euler equation (20) then pins down the response of z_B: as ℓ_a rises, $\Lambda(\ell_a, G)$ falls, so $\Omega_1 \Lambda(z_B, G)$ must increase to maintain the equality $\phi_z = \Omega_1 \Lambda(z_B, G) + \Omega_3 \Lambda(\ell_a, G)$, which requires z_B to fall. However, the z_B decline is bounded by the ℓ_{dc} increase, ensuring (50).

This liquidity increase affects the seller's intensive margin through the buyer's liquidity constraint. At a posted price ρ, the buyer is constrained when her available liquidity ℓ falls below the unconstrained expenditure level $\ell^*(\rho) \equiv \rho\, q^*(\rho)$. A constrained buyer spends all of ℓ, generating seller revenue $R(\rho, \ell) = \ell\,(1 - c/\rho)$; an unconstrained buyer purchases the efficient quantity, generating the strictly higher $R_{unc}(\rho) = (\rho - c)\, q^*(\rho)$. When i_e rises and total liquidity in CBDC-accepting meetings increases (Lemma 3), there are two effects at any given interior price ρ. First, banked buyers who remain constrained now carry more liquidity and therefore spend more, directly raising their contribution to $\bar{R}(\rho)$. Second, some banked buyers who were previously constrained at ρ now hold enough liquidity to cross the threshold $\ell^*(\rho)$ and become unconstrained, jumping from $R(\rho, \ell)$ to the higher $R_{unc}(\rho)$. Both effects raise $\bar{R}(\rho)$ at interior prices where CBDC-accepting buyers—who constitute the dominant share of the buyer population ($w_{dc} + w_a$)—are constrained.

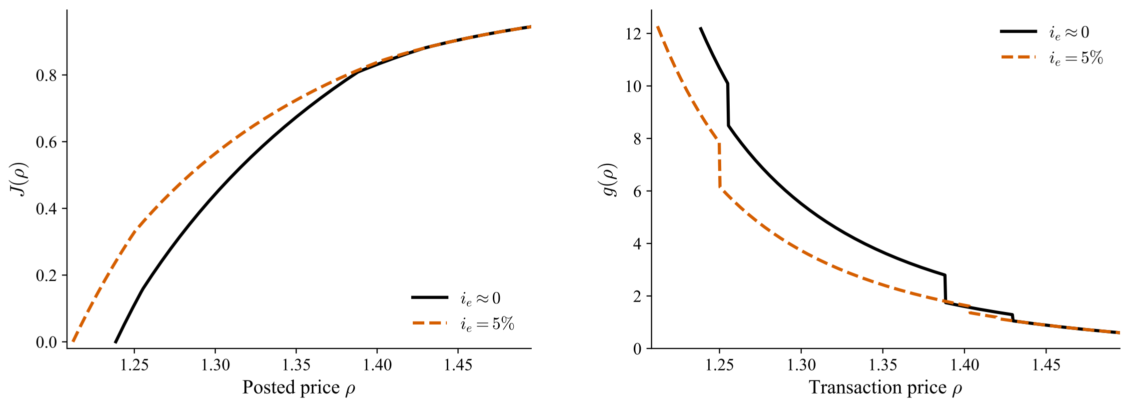
The next proposition formalizes this argument: a higher CBDC rate shifts the posted-price distribution toward lower prices in the sense of first-order stochastic dominance (FOSD).

PROPOSITION 3 (FOSD Price Shift). *Let $i_e^0 < i_e^1$ both exceed the entry threshold $\underline{i}_e$, and let J^0, J^1 denote the corresponding equilibrium posted-price CDFs. Then $J^1(\rho) \geq J^0(\rho)$ for all $\rho \in [\underline{\rho}, \bar{\rho}]$, with strict inequality on the interior. Equivalently, J^0 first-order stochastically dominates J^1: the price distribution shifts toward lower prices as i_e rises.*

Proof. See Appendix A.4. □

The equal-profit condition (Lemma 2) provides the mechanism. A seller posting a moderate price now earns higher per-customer revenue from the more liquid buyers, but every posted price must yield the same expected profit. The higher intensive margin at interior prices is offset by a lower extensive margin—more rival sellers crowd into the lower price region until the revenue gain is dissipated. The equilibrium response is a price distribution that places more mass at lower prices, reducing the average markup (Figure 1).

The term "positive price externality" captures the fact that this price shift benefits buyers who played no role in causing it. In the Burdett-Judd framework, sellers post a single price before observing whether the arriving buyer is banked or unbanked and which payment instruments are available. The price distribution therefore reflects the average buyer composition—it is shaped by $\bar{R}(\rho)$, which aggregates over all buyer states (29). When the CBDC rate rises, banked buyers carry more liquidity, raising $\bar{R}$ at interior prices and shifting the price distribution downward. The unbanked buyer, who holds no CBDC and does not interact with the banking sector, faces the same transaction price distribution G (7) as the banked buyer. She benefits from the lower prices without bearing any of the cost: her cash holdings z_U purchase more goods on average, raising her DM consumption and surplus.



A. Posted-price CDF $J(\rho)$

B. Transaction price density $g(\rho)$

FIGURE 1. Equilibrium price distribution at $i_e \approx 0$ (solid) and $i_e = 5\%$ (dashed). The CDF shifts leftward; $\underline{\rho}$ falls while $\bar{\rho}$ is nearly unchanged.

This channel is absent in models with competitive pricing or Nash bargaining, where the terms of trade depend only on the buyer's own portfolio. The Burdett-Judd price-posting framework, with its population-weighted $\bar{R}$, creates the cross-buyer spillover: the central bank's CBDC rate policy indirectly improves the terms of trade for the unbanked—a group that monetary policy cannot reach directly.

Moreover, the unbanked buyer amplifies the externality through her own portfolio response. Lower transaction prices mean that a given cash balance z_U purchases more goods, which tightens the liquidity constraint less frequently. The marginal liquidity benefit $\Lambda(z_U, G)$ falls, and the Euler equation requires a higher z_U to restore the equality $\phi_z = \Lambda(z_U, G)$. The unbanked buyer responds by accumulating more cash, which further raises her DM consumption and surplus.

Figure 1 illustrates this mechanism.[3] Panel A plots the posted-price CDF $J(\rho)$ at $i_e \approx 0$ (solid) and $i_e = 5\%$ (dashed). The price distribution shifts toward lower prices across the support: the lower bound $\underline{\rho}$ falls while the upper bound $\bar{\rho}$ is nearly unchanged, consistent with the prediction that the monopoly price depends on $\bar{R}$ only through the sufficient statistic ξ (Proposition 1), which varies little when the marginal buyer states are uncon-

[3]The figure uses illustrative parameters chosen to make the FOSD shift visually prominent: $\alpha_1 = 0.80$, $\Omega_2 = 0.55$, $\tau = 0.95$, with all other parameters at their baseline calibrated values (Table A2). The qualitative pattern—a leftward shift of $J(\rho)$ and a leftward shift with higher peak in $g(\rho)$—is robust across parameter configurations.

strained at $\bar{\rho}$. Panel B shows the corresponding transaction price density $g(\rho)$, which shifts leftward and becomes more concentrated at lower prices. The quantitative magnitude of this externality is assessed in Section 3.2.

3.2. Quantitative Results

We calibrate the model to the pre-CBDC U.S. economy and introduce CBDC as a counterfactual policy. The calibration targets four empirical moments—the money-to-output ratio, the aggregate markup, the deposit rate, and the net interest margin—using four internal parameters; the details are in Appendix B. We vary the CBDC rate i_e from 0 to 5% across three acceptance scenarios $\tau \in \{0.5, 0.7, 0.9\}$.

Portfolio rebalancing. The banked buyer holds $e_B > 0$ only when i_e exceeds a threshold $\underline{i}_e(\tau)$ at which the marginal liquidity benefit of the first CBDC unit equals its opportunity cost (Lemma 1). The threshold is decreasing in τ: wider CBDC acceptance raises the liquidity premium on CBDC balances, lowering the return needed to induce adoption. Even under near-universal acceptance, the CBDC rate must be well above zero before adoption occurs, because deposits already provide liquidity in all electronic meetings and CBDC is redundant in the $(1-\tau)$-fraction where it is not accepted.

Figure 2 traces the portfolio response. Once i_e crosses the threshold, CBDC balances grow rapidly (Panel D), while deposits decline partially (Panel C). The substitution is incomplete: deposits retain an exclusive role in deposit-only meetings (equation (48)), so the buyer optimally holds both instruments. Panels E and F show that the CBDC inflow more than offsets the deposit outflow: total banked holdings $z_B + d_B + e_B$ and effective liquidity $\ell_a = z_B + (1 + i_d)d_B + (1 + i_e)e_B$ both rise, confirming Lemma 3 quantitatively.

The response of the unbanked buyer is particularly revealing. Panel A shows that z_U increases above the entry threshold, even though the unbanked buyer holds no CBDC and has no bank account. This is the first quantitative signature of the positive price externality: as the equilibrium price distribution shifts downward (Proposition 3), a given cash balance purchases more goods, which relaxes the liquidity constraint. The unbanked buyer's Euler equation then requires higher cash holdings to restore the equality between the marginal cost of carrying cash and the marginal liquidity benefit.

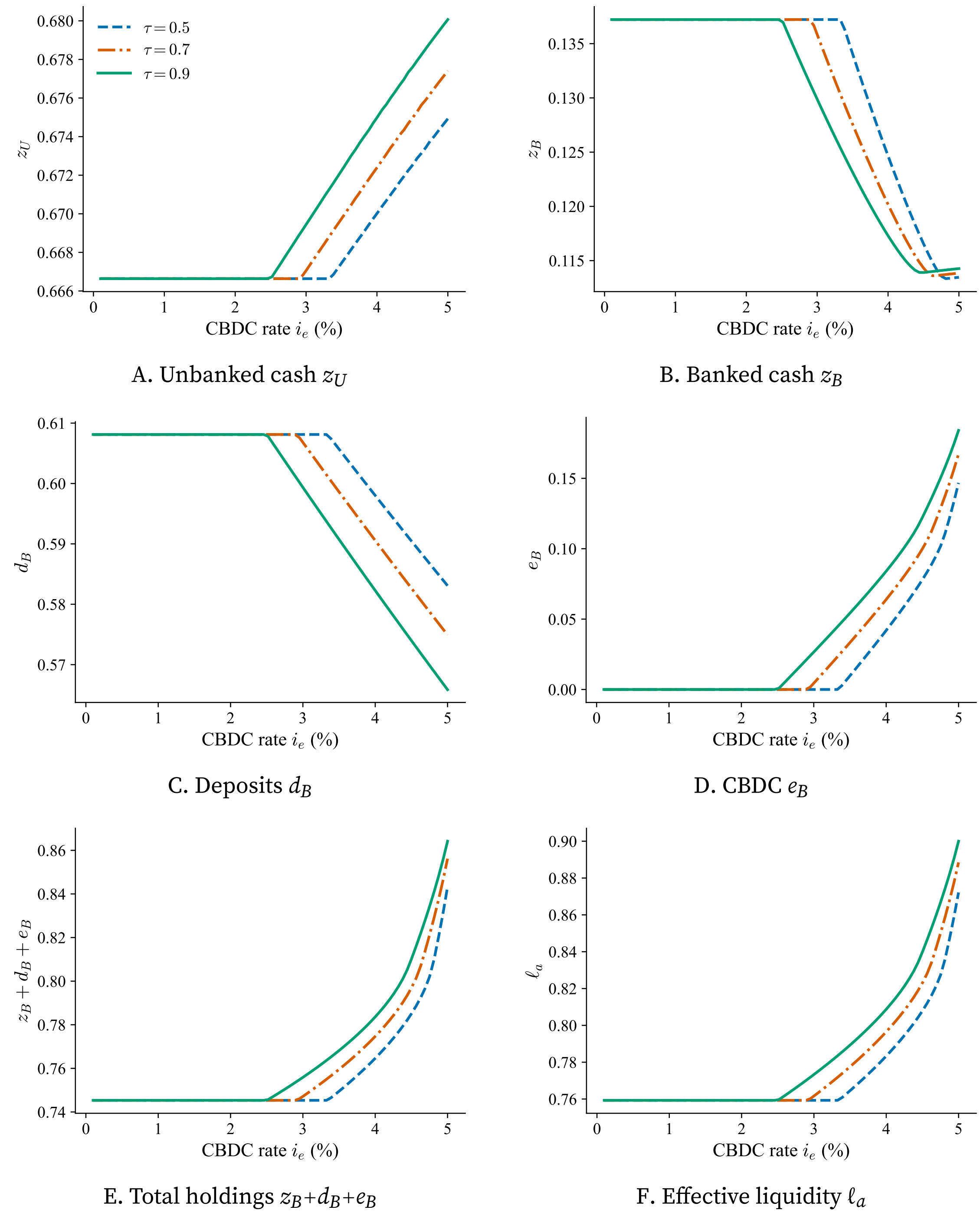
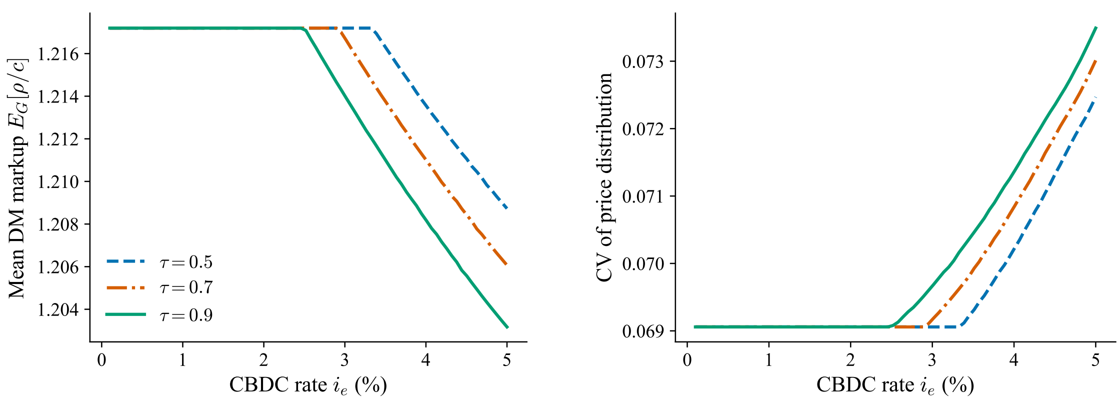


FIGURE 2. Portfolio allocations as i_e varies. Deposits decline and CBDC balances grow above the entry threshold; z_U rises via the positive price externality; effective liquidity ℓ_a rises overall.

Positive price externality. Figure 3 confirms the central theoretical prediction of Section 3.1. Panel A shows that the mean DM markup $E_G[\rho/c]$ declines monotonically as i_e rises above the entry threshold. The decline is steeper under wider acceptance: a higher τ amplifies the banked buyer's portfolio rebalancing, which raises the expected intensive margin $\bar{R}(\rho)$ at interior prices and, through the equal-profit condition, compresses the entire price distribution downward—precisely the FOSD shift of Proposition 3. Panel B shows that price dispersion, measured by the coefficient of variation of the transaction price distribution, also falls. As the price support narrows from below ($\underline{\rho}$ falls while $\bar{\rho}$ is nearly unchanged), the distribution becomes more concentrated at lower prices.

The markup decline is the mechanism through which CBDC liquidity reaches non-holders. In the Burdett-Judd framework, sellers post prices without observing whether the arriving buyer is banked or unbanked. The posted-price distribution therefore reflects the average buyer composition through $\bar{R}(\rho)$. When banked buyers carry more liquidity, the equal-profit condition forces sellers to shift mass toward lower prices, and the unbanked buyer—who played no role in causing this shift—faces the same, now lower, price distribution.



A. Mean DM markup $E_G[\rho/c]$ B. Price dispersion (CV)

FIGURE 3. DM markup response. Mean markup and price dispersion both fall as i_e rises.

Welfare effects. We measure welfare by the expected DM total surplus, following the standard convention in the New Monetarist literature (Chiu et al. 2023; Jiang and Zhu 2024). With quasi-linear CM preferences, the welfare comparison across steady states reduces to comparing DM surpluses: the CM labor adjustment fully absorbs changes in portfolio costs, lump-sum transfers T, and bank dividends Π, so these terms cancel across steady states. Entrepreneur net output $f(L) - R_l L$ is redistributed to banks as loan repayment and ultimately to households as dividends, and therefore also cancels under quasi-linearity. The measure excludes the bank-account maintenance cost ε (relevant only in Section 4, where λ is endogenized) and CBDC adoption costs κ (relevant only in Section 5). When these costs matter, we discuss their welfare implications separately.[4] Aggregate welfare decomposes into type-specific contributions:

$$W = (1-\lambda)\, W_U + \lambda\, W_B, \tag{51}$$

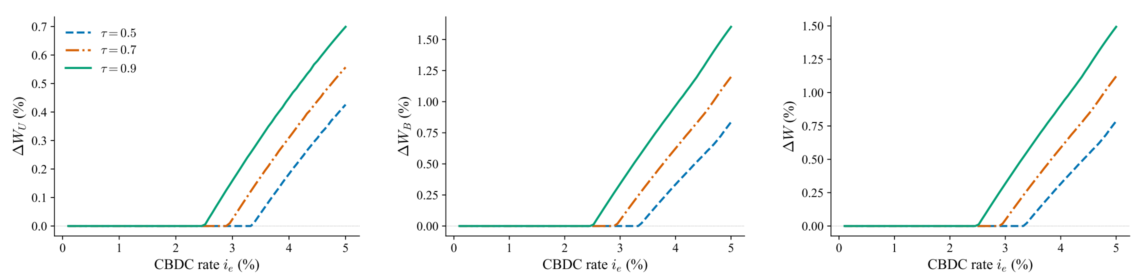
where the unbanked surplus is $W_U = \int [u(q_U(\rho)) - c\, q_U(\rho)]\, dG(\rho)$ and the banked surplus aggregates across four meeting types:

$$W_B = \Omega_1 S_m + \Omega_2 (1-\tau)\, S_d + \Omega_2 \tau\, S_{dc} + \Omega_3 S_a, \tag{52}$$

with $S_j = \int [u(q_j(\rho)) - c\, q_j(\rho)]\, dG(\rho)$ denoting the expected total surplus in meeting type j.

Figure 4 plots the welfare changes relative to the no-CBDC baseline. The central result is in Panel A: the unbanked welfare gain ΔW_U is strictly positive above the entry threshold and monotonically increasing in τ. The unbanked buyer holds no CBDC, has no bank account, and does not interact with the banking sector—yet she benefits from the CBDC policy purely through the lower equilibrium price distribution. This is the welfare manifestation of the positive price externality.

Panel B shows that the banked buyer's welfare gain ΔW_B is also positive and larger in magnitude. The banked buyer benefits directly through interest-bearing CBDC, which provides additional purchasing power in $\Omega_2\tau$ and Ω_3 meetings, and indirectly through the same price distribution shift that benefits the unbanked. Panel C confirms that aggregate welfare ΔW is positive across all acceptance scenarios, with gains comparable in magnitude to those reported in Chiu et al. (2023) and Jiang and Zhu (2024) for calibrated New Monetarist CBDC models.



A. Unbanked ΔW_U B. Banked ΔW_B C. Aggregate ΔW

FIGURE 4. Welfare changes relative to the no-CBDC baseline. All measures are positive above the entry threshold and increasing in τ.

[4] An alternative would be to include all CM-side costs in a single welfare criterion. Under quasi-linearity, doing so would add a constant (the aggregate of ε and κ costs at the endogenous λ and τ) without changing the ranking of CBDC policies for a given (λ, τ). We follow the DM-surplus convention to maintain comparability with Chiu et al. (2023) and Jiang and Zhu (2024).

4. Endogenous Financial Inclusion

The preceding analysis treats the banked fraction λ as exogenous. Because interest-bearing CBDC is available only to banked buyers, a higher CBDC rate directly raises the net value of maintaining a bank account. This section endogenizes the financial inclusion decision: Section 4.1 models the buyer's choice to open a bank account as a cost–benefit comparison, and Section 4.2 shows quantitatively that CBDC can drive near-universal financial inclusion under the calibrated U.S. economy.

4.1. Endogenous Financial Inclusion

The analysis in Section 3 treats the banked fraction λ as exogenous. In practice, a household's decision to open and maintain a bank account reflects a cost–benefit comparison that CBDC directly alters. Interest-bearing CBDC is available only to banked buyers: it expands the banked buyer's asset menu without affecting the unbanked buyer's portfolio choice. When the CBDC rate rises, the banked buyer optimally holds a positive CBDC balance that earns interest and provides additional purchasing power in CBDC-accepting meetings, raising the banked buyer's expected DM surplus. The unbanked buyer, who holds only cash, receives no such direct benefit. The net value of maintaining a bank account therefore increases with the CBDC rate, and marginal unbanked buyers—those whose banking costs barely exceed the pre-CBDC net benefit—find it worthwhile to open accounts. This section endogenizes λ and shows that the resulting financial inclusion amplifies the positive price externality documented in Section 3.

To endogenize financial inclusion, we introduce heterogeneity in the per-period cost of maintaining a bank account. Each buyer draws a banking cost ε from a cumulative distribution function F with support $[0, \bar{\varepsilon}]$, where $F(0) = 0$ and $F(\bar{\varepsilon}) = 1$. The cost ε, denominated in CM utility, captures all frictions associated with banking: account maintenance fees, the opportunity cost of meeting minimum balance requirements, and transportation costs.[5] From the CM problem (8), quasi-linearity implies that the portfolio choice is independent of wealth. The net value of the optimal portfolio—the discounted expected DM surplus minus the inflation cost of carrying assets—is

$$\Psi_B \equiv \max_{z_B, d_B, e_B} \left\{ -\gamma(z_B + d_B + e_B) + \beta V^{bk}(z_B, d_B, e_B) \right\}, \tag{53}$$

$$\Psi_U \equiv \max_{z_U} \left\{ -\gamma z_U + \beta V^U(z_U) \right\}. \tag{54}$$

[5] The 2019 FDIC Survey of Household Use of Banking and Financial Services reports that the most frequently cited reasons for being unbanked include minimum balance requirements, distrust of banks, and unpredictable fees. We abstract from these heterogeneous motives by summarizing them as a single cost parameter.

The CM consumption surplus $U(x^*) - x^*$ and the wealth terms (lump-sum transfers T and bank dividends Π) are identical for both buyer types and therefore cancel from the banking decision. A buyer with cost ε opens a bank account if and only if $\Psi_B - \Psi_U \geq \varepsilon$.

Define the threshold banking cost

$$\tilde{\varepsilon} \equiv \Psi_B - \Psi_U. \tag{55}$$

All buyers with $\varepsilon \leq \tilde{\varepsilon}$ choose to be banked, so the equilibrium banked fraction is

$$\lambda^* = F(\tilde{\varepsilon}) = F(\Psi_B - \Psi_U)\,. \tag{56}$$

The key comparative static is immediate. Raising i_e increases Ψ_B, because the banked buyer gains access to a higher-return asset that generates additional DM liquidity in $\Omega_2\tau$ and Ω_3 meetings. The unbanked buyer's surplus Ψ_U changes only indirectly, through the equilibrium price distribution that shifts downward under the positive price externality. The direct increase in Ψ_B dominates, so the threshold $\tilde{\varepsilon}$ rises with i_e, pulling marginal unbanked buyers into the financial system.

The equilibrium involves a fixed point: the portfolios $(z_U^*, z_B^*, d_B^*, e_B^*)$ and the price distribution G^* depend on λ through the expected intensive margin $\bar{R}$ (equation (29)), while λ in turn depends on these equilibrium objects through $\Psi_B - \Psi_U$. Formally, a *stationary monetary equilibrium with endogenous financial inclusion* augments Definition 1 by replacing the exogenous λ with the fixed-point condition (56).

PROPOSITION 4 (Existence with endogenous λ). *Under the maintained assumptions and F continuous with $F(0) = 0$, an equilibrium with endogenous financial inclusion exists: there exists $\lambda^* \in [0, 1]$ satisfying* (56).

Proof. See Appendix A.11. The mapping $\Phi(\lambda) = F(\Psi_B(\lambda) - \Psi_U(\lambda))$ is a continuous self-map of $[0, 1]$; existence follows from Brouwer's fixed-point theorem. □

4.2. Quantitative Analysis

We use the calibrated parameters from Section 3.2 and specialize the banking-cost distribution to the log-normal case $F(\varepsilon; \mu, \sigma) = \Phi((\ln \varepsilon - \mu)/\sigma)$. The two parameters (μ, σ) are pinned by two conditions. First, $\sigma = 0.5866$ is estimated directly from data: treating $\varepsilon = 1/(\text{branch density per capita})$ as a proxy for the banking access cost following Lahcen and Gomis-Porqueras (2021), we fit a log-normal distribution to county-level branch density from the FDIC Summary of Deposits (2011) across $N = 3{,}188$ US counties via maximum likelihood, which yields $\hat{\sigma} = 0.5866$. Second, μ is pinned by requiring that the endogenous banked fraction equals the pre-CBDC observed rate at $i_e = 0$: $F(\tilde{\varepsilon}_0; \mu, \sigma) = \lambda_0 = 0.88$, where $\tilde{\varepsilon}_0 = \Psi_B(i_e = 0) - \Psi_U(i_e = 0)$ is the baseline inclusion threshold.[6] The results are displayed in Figures 5–7. In all panels, the three lines correspond to CBDC acceptance rates $\tau \in \{0.5, 0.7, 0.9\}$.

Financial inclusion. Figure 5 plots the equilibrium banked fraction λ^* as a function of the CBDC rate. Under all three acceptance rates the economy starts at $\lambda^* = 0.88$ and remains at this level until the CBDC rate crosses the adoption threshold $\underline{i}_e$. Above the threshold, λ^* rises sharply toward unity. A more widely accepted CBDC triggers financial inclusion at a lower rate because a higher τ expands the fraction of DM meetings in which CBDC provides additional purchasing power, raising the banked buyer's DM surplus Ψ_B and therefore the threshold banking cost $\tilde{\varepsilon}$.

The mechanism is the direct enrichment of the banked buyer's asset menu identified in Section 4.1. When i_e is high enough that CBDC enters the banked buyer's optimal portfolio, the CBDC balance earns interest and provides additional liquidity in $\Omega_2\tau$ and Ω_3 meetings. This raises Ψ_B relative to Ψ_U, widening the net benefit of banking and pulling marginal unbanked buyers into the financial system. The inclusion response is sharp once the adoption threshold is crossed: moderate CBDC rates can trigger near-universal banking.

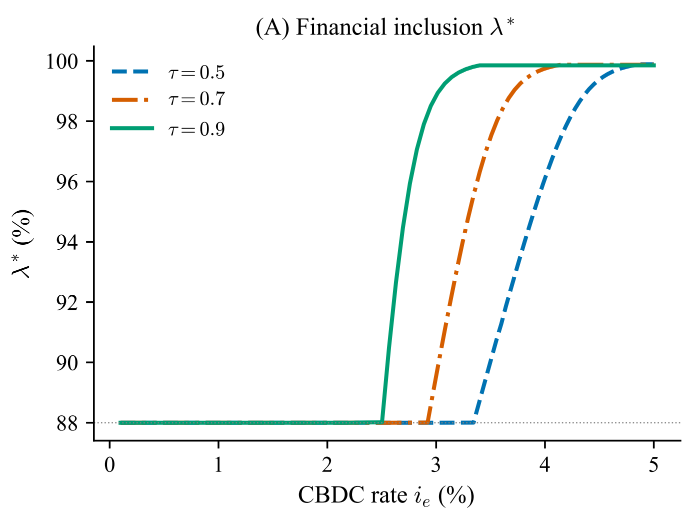


FIGURE 5. Banked fraction λ^* as i_e varies ($\lambda_0 = 0.88$, log-normal $F(\varepsilon)$, FDIC calibration). Above the adoption threshold, λ^* rises sharply toward unity.

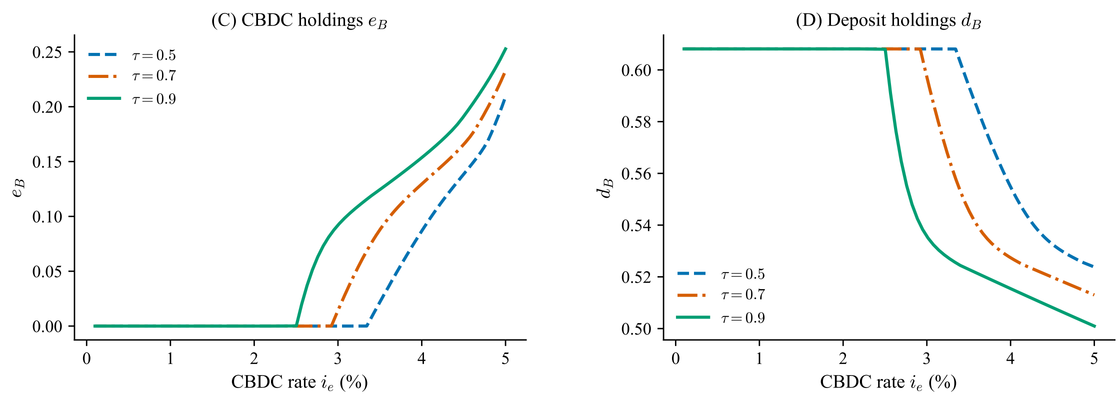
Portfolio dynamics. Figure 6 shows the portfolio and rate responses under endogenous inclusion. Panel (a) shows that banked buyers accumulate CBDC balances e_B as i_e rises

[6]The baseline threshold is $\tilde{\varepsilon}_0 = 0.1258$, giving $\mu = -2.762$. The implied conditional banking cost is $\kappa \equiv \mathbb{E}[\varepsilon \mid \varepsilon \le \tilde{\varepsilon}_0]/W_0 = 10.1\%$, where W_0 is the baseline buyer surplus; this is a data-determined output, not a calibration target.

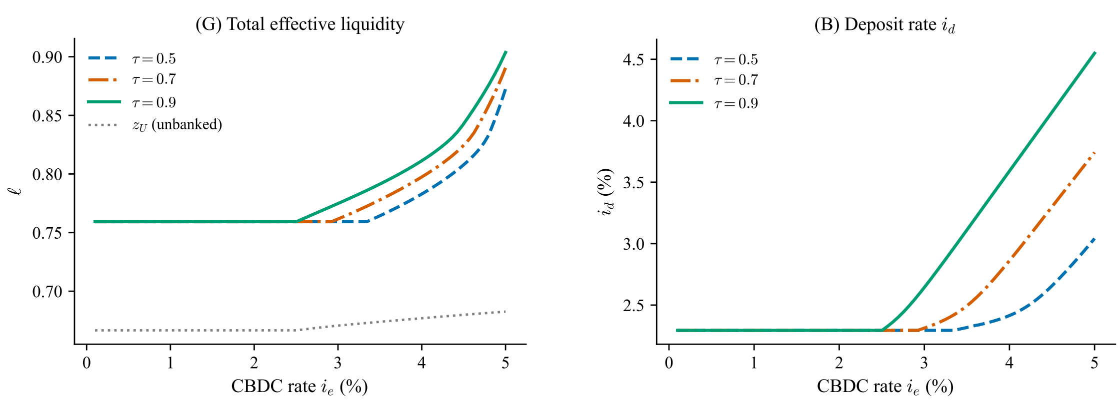
above the adoption threshold. Panel (b) shows that deposit holdings d_B decline partially: CBDC displaces deposits in the banked buyer's portfolio as the two instruments compete for the same DM purchasing role in CBDC-accepting meetings. Despite this decline, total effective liquidity $\ell_B \equiv z_B + (1+i_d)d_B + (1+i_e)e_B$ rises unambiguously (Panel (c)): the CBDC balance more than offsets the deposit outflow, expanding the effective purchasing power of banked buyers in the DM. Cash holdings of unbanked buyers z_U remain nearly flat, as unbanked buyers are affected only indirectly through the endogenous price distribution. Panel (d) shows that the deposit rate i_d rises as banks adjust to the changing portfolio composition.

Positive price externality under endogenous inclusion. Figure 7 shows the markup dynamics. Panel (a) reports the mean DM markup, which falls monotonically once CBDC enters the economy. The mechanism is the same positive price externality identified in Section 3: CBDC provides additional DM liquidity to banked buyers, which tightens the price support through the equal-profit condition and compresses the equilibrium markup distribution. Under endogenous λ, this effect is amplified by a composition channel: as λ^* approaches unity, the seller's expected revenue function $\bar{R}(\rho)$ is increasingly shaped by the banked buyer's demand elasticity, further compressing prices. Panel (b) shows that price dispersion, measured by the markup coefficient of variation, rises once financial inclusion begins. This reflects the entry of newly banked buyers with heterogeneous liquidity levels into the market, which widens the effective buyer pool and expands the price support. The dispersion effect runs counter to the mean markup effect: while the average markup falls, the spread of prices widens as the market accommodates a more heterogeneous buyer population.

Two implications for CBDC design follow. First, the financial inclusion channel requires that CBDC be accessible through the banking system: a fully anonymous, cash-like CBDC available to all buyers regardless of banking status would not raise Ψ_B relative to Ψ_U and would generate no inclusion incentive. Second, both financial inclusion and markup compression are monotone in acceptance: a central bank should prioritize building a wide acceptance network because a higher τ delivers the same inclusion and price gains at a lower required CBDC rate.

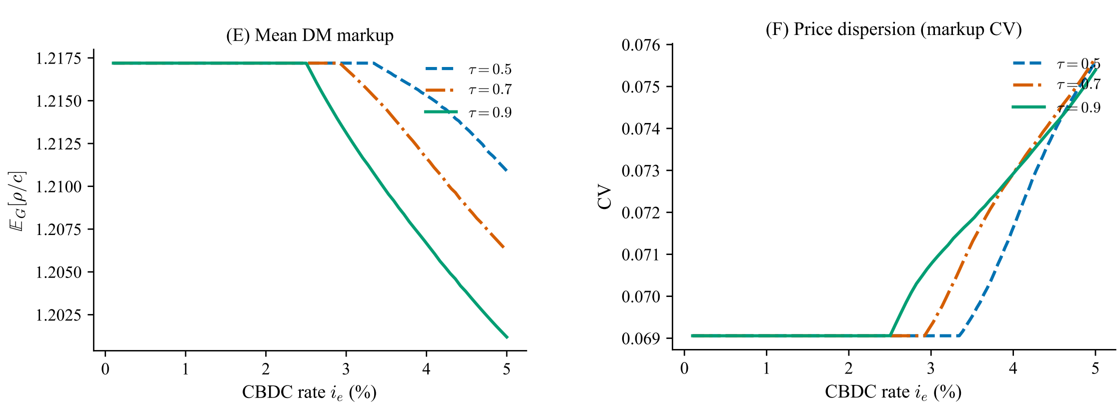


A. CBDC holdings e_B B. Deposit holdings d_B



C. Total effective liquidity ℓ_B and z_U D. Deposit rate i_d

FIGURE 6. Portfolio dynamics and deposit rate under endogenous financial inclusion. CBDC balances accumulate above the threshold; deposits decline partially; total liquidity ℓ_B rises; the deposit rate adjusts to the changing portfolio composition.



A. Mean markup $\mathbb{E}_G[\rho/c]$ B. Markup coefficient of variation

FIGURE 7. Markup dynamics under endogenous financial inclusion. Mean markup falls (positive price externality amplified by composition); markup CV rises as newly banked buyers widen the buyer pool.

5. Endogenous CBDC Acceptance

The previous two sections treat the CBDC acceptance rate τ as exogenous. In practice, sellers choose whether to bear the cost of accepting CBDC based on the expected revenue gain. Higher CBDC rates raise buyer liquidity in CBDC-accepting meetings, generating a positive revenue gain for sellers that can justify adoption costs. This section endogenizes τ: Section 5.1 characterizes the seller's adoption decision and equilibrium structure, and Section 5.2 evaluates the model quantitatively.

5.1. Seller's Adoption Decision and Equilibrium

The analysis so far treats the CBDC acceptance rate τ as exogenous. In practice, accepting CBDC requires sellers to upgrade payment terminals, train staff, and bear ongoing network fees—costs that must be justified by additional revenue. This section endogenizes τ by modeling the seller's adoption decision as a cost–benefit comparison, following the approach of Gong, Han, and Zhu (2023). Each seller is a measure-zero agent in a continuum. An individual adoption decision therefore does not affect the aggregate acceptance rate τ, the buyer's portfolio, or the price distribution $J(\rho)$—all of which the seller takes as given. The equilibrium acceptance rate τ^* is determined as a fixed point: at τ^*, the mass of sellers whose revenue gain exceeds their adoption cost equals τ^*.

Revenue gain from CBDC acceptance. Each seller j draws an adoption cost κ_j from a cumulative distribution function H with support $[0, \bar{\kappa}]$, where $H(0) = 0$ and $H(\bar{\kappa}) = 1$. The cost κ_j, denominated in CM utility, captures the per-period expense of maintaining CBDC payment infrastructure.[7] Sellers are otherwise identical: they share the same production technology and face the same buyer population.

The meeting types $\Omega_1, \Omega_2, \Omega_3$ classify the payment infrastructure available in a given encounter. A seller who invests in CBDC payment technology can process CBDC in any meeting where the infrastructure supports it; a non-adopting seller cannot. In electronic meetings (Ω_2), a non-accepting seller faces buyer liquidity $\ell_d = (1 + i_d)d$ (deposits only), while an accepting seller captures $\ell_{dc} = (1+i_d)d+(1+i_e)e$. In all-instrument meetings (Ω_3), all three payment channels—cash, deposits, and CBDC—are technologically available. However, processing a CBDC payment still requires the seller's own infrastructure. An adopting seller in an Ω_3 meeting faces the full buyer liquidity $\ell_a = z + (1 + i_d)d + (1 + i_e)e$; a non-adopting seller can only accept cash and deposits, so the buyer's available liquidity is

$$\ell_{md} \equiv z + (1 + i_d)\, d, \tag{57}$$

[7] The heterogeneous adoption cost can also be interpreted as reflecting variation in sellers' technological readiness or customer composition.

where the subscript *md* denotes "money and deposits."[8] In cash-only meetings (Ω_1) and for unbanked buyers, the two seller types generate identical revenue.

A seller's meeting-type probabilities depend on the buyer population and the matching technology, not on other sellers' adoption decisions. Conditional on meeting a banked buyer, the probability of an electronic meeting is $\lambda\Omega_2$ and that of an all-instrument meeting is $\lambda\Omega_3$, regardless of the aggregate acceptance rate τ. Accordingly, the non-accepting and accepting sellers' expected intensive margins from banked buyers are[9]

$$\bar{R}_N(\rho) = \lambda\Omega_2 R_d(\rho) + \lambda\Omega_3 R_{md}(\rho), \tag{58}$$

$$\bar{R}_A(\rho) = \lambda\Omega_2 R_{dc}(\rho) + \lambda\Omega_3 R_a(\rho), \tag{59}$$

where $R_j(\rho)$ denotes per-customer revenue when the buyer has liquidity ℓ_j. The per-customer revenue gain from CBDC acceptance is

$$\Delta R(\rho) \equiv \bar{R}_A(\rho) - \bar{R}_N(\rho) = \lambda\Omega_2\big[R_{dc}(\rho) - R_d(\rho)\big] + \lambda\Omega_3\big[R_a(\rho) - R_{md}(\rho)\big]. \tag{60}$$

The revenue gain has two channels. The *electronic-meeting channel* (Ω_2 term) captures the gain from processing CBDC in meetings where only electronic instruments are available: the adopting seller accesses ℓ_{dc} instead of ℓ_d, raising revenue by $(1+i_e)e(1-c/\rho)$ in the constrained region. The *all-instrument channel* (Ω_3 term) captures the gain from processing CBDC in meetings where cash is also available: the adopting seller accesses ℓ_a instead of ℓ_{md}, with the same liquidity increment $(1+i_e)e$. Crucially, the Ω_3 channel operates even at $\tau = 0$: because buyers may hold positive CBDC balances for use in all-instrument meetings (via the Ω_3 term in the CBDC Euler equation (A4)), a deviating seller who adopts CBDC captures this liquidity premium regardless of other sellers' behavior.

LEMMA 4. *If $e > 0$ in equilibrium, then $\Delta R(\rho) > 0$ for all $\rho \in [\underline{\rho}, \bar{\rho})$, and $\Delta R(\bar{\rho}) \geq 0$.*

Proof. See Appendix A.6. □

The lemma requires only $e > 0$—no restriction on τ. The economic logic is straightforward: CBDC holdings raise the buyer's available liquidity at adopting sellers in both Ω_2 and Ω_3 meetings. At any price ρ where the buyer is liquidity-constrained without CBDC, the additional balance increases expenditure and hence seller revenue. At the monopoly price $\bar{\rho}$, the buyer may already be unconstrained, yielding $\Delta R(\bar{\rho}) = 0$. Since binding constraints obtain on the interior when $e > 0$ (the maintained assumption $\hat{\rho}_{dc} \leq \bar{\rho}$ from Proposition 1), the gain is strictly positive on $[\underline{\rho}, \bar{\rho})$.

[8] The aggregate $\bar{R}(\rho;\tau)$ used for the equal-profit condition in Section 2 treats all sellers in Ω_3 meetings symmetrically. This simplification is innocuous for equilibrium computation (since $(1+i_e)e$ is small relative to ℓ_a in calibrated equilibria) but matters for the *differential* revenue that drives adoption incentives.

[9] Equations (58)–(59) omit the unbanked and cash-only terms, which are identical for both seller types and cancel in the revenue difference.

Both seller types post from the same price distribution $J(\rho)$, pinned by the equal-profit condition applied to the aggregate intensive margin $\bar{R}(\rho;\tau)$ from Section 2.[10] The expected profit gain from acceptance is

$$\Delta\bar{\Pi}(\tau) \equiv \int_{\underline{\rho}}^{\bar{\rho}} \left[\alpha_1 + 2\alpha_2(1 - J(\rho))\right] \Delta R(\rho)\, dJ(\rho) > 0 \quad \text{whenever } e > 0. \tag{61}$$

A higher acceptance rate τ amplifies $\Delta\bar{\Pi}$ through the *portfolio channel*: the CBDC Euler equation (A4) contains the term $\Omega_2\tau\Lambda(\ell_{dc})$, so higher τ lowers the effective holding cost of CBDC, inducing buyers to hold more. Larger CBDC balances widen both $R_{dc} - R_d$ and $R_a - R_{md}$, raising $\Delta R(\rho)$ at every ρ. The Ω_3 channel provides a baseline: even at $\tau = 0$, CBDC holdings from the $\Omega_3\Lambda(\ell_a)$ term in the Euler equation generate positive ΔR whenever $e > 0$.

Adoption equilibrium. Seller j adopts if and only if $\Delta\bar{\Pi}(\tau) \geq \kappa_j$, yielding the fixed-point condition

$$\tau^* = H\big(\Delta\bar{\Pi}(\tau^*)\big). \tag{62}$$

An equilibrium with endogenous acceptance augments Definition 1 by replacing the exogenous τ with this fixed-point condition.

PROPOSITION 5. *Let $\underline{i}_e(0)$ denote the portfolio entry threshold from Lemma 1 evaluated at $\tau = 0$. For any H with $H(0) = 0$:*

(i) *If $i_e \leq \underline{i}_e(0)$, then $e = 0$ in the $\tau = 0$ equilibrium, and $\tau^* = 0$ is a stationary monetary equilibrium.*

(ii) *If $i_e > \underline{i}_e(0)$, then $e > 0$ in the $\tau = 0$ equilibrium, $\Delta\bar{\Pi}(0) > 0$, and $\tau^* = 0$ is* not *an equilibrium. There exists $\tau^* > 0$ satisfying* (62).

Proof. See Appendix A.8. □

Part (ii) is the key structural implication of the Ω_3 channel. When the CBDC rate exceeds the portfolio entry threshold, buyers hold positive CBDC balances for use in all-instrument meetings, regardless of seller adoption. These CBDC holdings create a revenue gap between adopting and non-adopting sellers in Ω_3 meetings, breaking the coordination failure that sustains the no-adoption equilibrium. The Ω_3 channel thus provides a built-in "bootstrap" mechanism: the central bank need not engineer a simultaneous jump in buyer CBDC demand and seller acceptance—positive CBDC holdings by buyers automatically

[10] An alternative approach assigns separate price distributions to accepting and non-accepting sellers. The two-distribution equilibrium is significantly more complex and does not qualitatively change the adoption incentives. We adopt the single-distribution approach for tractability, following Gong, Han, and Zhu (2023).

trigger adoption by the lowest-cost sellers, initiating the feedback loop between buyer portfolios and seller acceptance.

PROPOSITION 6. *Fix $i_e > \underline{i}_e(0)$ and suppose $\gamma > \beta$. The profit gain function satisfies:*

(i) $\Delta\bar{\Pi}(0) \geq 0$, *with* $\Delta\bar{\Pi}(0) = 0$ *if and only if* $e = 0$ *in the* $\tau = 0$ *equilibrium (equivalently,* $i_e \leq \underline{i}_e(0)$*).*

(ii) $\Delta\bar{\Pi}(\tau) > 0$ *and* $\Delta\bar{\Pi}'(\tau) > 0$ *for all* $\tau \in (0,1)$ *at which* $e > 0$.

(iii) $\Delta\bar{\Pi}(\tau)$ *is continuous on* $(0,1)$.

Proof. See Appendix A.7. □

The monotonicity of $\Delta\bar{\Pi}$ in τ operates entirely through the intensive margin. Unlike the unconditional-weight formulation where the meeting base $w_{dc} = \lambda\Omega_2\tau$ expands linearly with τ, the conditional weights $\lambda\Omega_2$ and $\lambda\Omega_3$ are independent of τ. Instead, higher τ raises CBDC balances e through the portfolio channel (equation (A4)), widening the per-meeting revenue gaps $R_{dc} - R_d$ and $R_a - R_{md}$, which enter ΔR multiplicatively. Economically, when more sellers accept CBDC, buyers hold more CBDC, and each adopting seller earns a larger revenue premium.

Specialize to the uniform distribution $H(\kappa) = \kappa/\bar{\kappa}$, so the fixed-point condition becomes

$$\tau^* = \min\left\{\frac{\Delta\bar{\Pi}(\tau^*)}{\bar{\kappa}},\, 1\right\}. \tag{63}$$

PROPOSITION 7. *Suppose $H(\kappa) = \kappa/\bar{\kappa}$ and define $\Gamma(\tau) \equiv \min\{\Delta\bar{\Pi}(\tau)/\bar{\kappa}, 1\}$.*

(i) *If* $i_e \leq \underline{i}_e(0)$ *(so* $\Delta\bar{\Pi}(0) = 0$*):*

  (a) *If* $\Delta\bar{\Pi}'(0^+)/\bar{\kappa} \leq 1$, *then* $\tau^* = 0$ *is the unique equilibrium (under the maintained assumption that* $\Delta\bar{\Pi}(\tau)/\tau$ *is non-increasing).*

  (b) *If* $\Delta\bar{\Pi}'(0^+)/\bar{\kappa} > 1$, *then at least one stable positive-adoption equilibrium* $\tau_H^* \in (0,1]$ *exists alongside* $\tau^* = 0$. *The critical cost* $\bar{\kappa}^* \equiv \Delta\bar{\Pi}'(0^+)$ *is the bifurcation boundary.*

(ii) *If* $i_e > \underline{i}_e(0)$ *(so* $\Delta\bar{\Pi}(0) > 0$*):* $\tau^* = 0$ *is not an equilibrium. There exists at least one* $\tau^* > 0$ *satisfying* $\Gamma(\tau^*) = \tau^*$. *The equilibrium* τ^* *is decreasing in* $\bar{\kappa}$*: higher adoption costs reduce the equilibrium acceptance rate.*

Uniqueness of the positive fixed point in both cases is confirmed numerically across all calibrated parameter values.

Proof. See Appendix A.9. □

The CBDC rate i_e is the central bank's primary instrument for activating seller adoption, operating through two distinct regimes. When $i_e \leq \underline{i}_e(0)$, buyers hold no CBDC at $\tau = 0$, and the economy faces a coordination failure: no seller adopts because no other seller has adopted. Breaking this trap requires the central bank to set i_e high enough to push $\bar{\kappa}$ below the critical threshold $\bar{\kappa}^*$, enabling the positive-adoption equilibrium. When $i_e > \underline{i}_e(0)$, the Ω_3 channel eliminates the coordination failure entirely. Buyers hold positive CBDC balances for use in all-instrument meetings, which creates immediate revenue gains for adopting sellers. In this regime, some positive adoption occurs for any cost distribution, and the policy question shifts from *whether* adoption occurs to *how much*: higher i_e raises τ^* by increasing CBDC balances and widening the revenue gap.

When positive adoption obtains, the endogenous acceptance rate τ^* is jointly determined with the equilibrium portfolios and price distribution. The positive price externality of Section 3 then operates at the endogenous τ^*: as buyers hold more CBDC under wider acceptance, their effective liquidity rises, compressing the equilibrium markup distribution and benefiting all buyers—including the unbanked—through lower transaction prices.

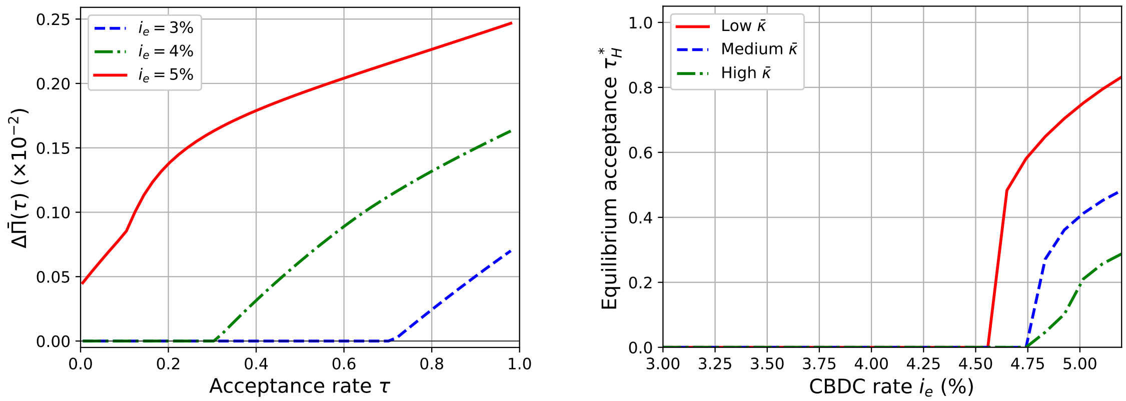
5.2. Quantitative Analysis

This subsection evaluates the endogenous CBDC acceptance model quantitatively. The adoption cost distribution is set to $H(\kappa) = \kappa/\bar{\kappa}$ (uniform on $[0, \bar{\kappa}]$), and all remaining parameters are identical to the baseline calibration of Section 3.2. The calibration of $\bar{\kappa}$ and the cost scenario definitions are detailed in Appendix C. We report results for three scenarios—low cost ($\bar{\kappa} = 1.2\,\bar{\kappa}^*$), medium cost ($\bar{\kappa} = 1.8\,\bar{\kappa}^*$), and high cost ($\bar{\kappa} = 2.8\,\bar{\kappa}^*$)—where $\bar{\kappa}^* \equiv \max_\tau \Delta\bar{\Pi}(\tau)$ at $i_e = 5\%$.

Figure 8 presents the two key objects of the endogenous acceptance model. Panel A plots $\Delta\bar{\Pi}(\tau)$ for three CBDC rates. Consistent with Proposition 6, the function is monotonically increasing in τ. The figure illustrates the two regimes identified in Proposition 7. At $i_e = 3\%$ and 4%, $\Delta\bar{\Pi}(0) = 0$ (Regime A): buyers hold no CBDC when no seller accepts it, and the profit gain emerges only after τ exceeds a threshold at which CBDC enters the portfolio. At $i_e = 5\%$, $\Delta\bar{\Pi}(0) > 0$ (Regime B): buyers hold CBDC for use in Ω_3 meetings regardless of seller adoption, so the revenue advantage is strictly positive even at $\tau = 0$. This confirms the Ω_3 bootstrap mechanism of Proposition 5(ii).

Panel B traces the equilibrium acceptance rate τ_H^* as a function of i_e for the three cost scenarios. Below the portfolio threshold $\underline{i}_e(0)$, no CBDC is held at $\tau = 0$ and the coordination failure persists: $\tau_H^* = 0$. Once i_e exceeds $\underline{i}_e(0)$, the Ω_3 channel activates and τ_H^* rises sharply. Lower adoption costs shift the activation point to lower i_e and produce higher equilibrium acceptance rates at each i_e, as the revenue gain needs to cover a smaller cost to justify adoption.

Figure 9 reports macroeconomic outcomes at the endogenous τ^* under the low-cost scenario. Below the portfolio threshold $\underline{i}_e(0) \approx 4.5\%$, the economy remains in the no-adoption equilibrium: $\tau^* = 0$, no CBDC is held, and all variables equal their no-CBDC baseline values. Once i_e crosses this threshold, adoption activates through the Ω_3 bootstrap, and the acceptance rate rises sharply (Panel A). This triggers the same portfolio



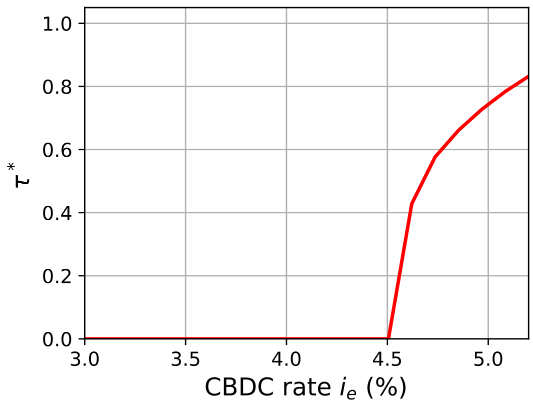
A. Profit gain $\Delta\bar{\Pi}(\tau)$

B. Equilibrium acceptance τ_H^* vs i_e

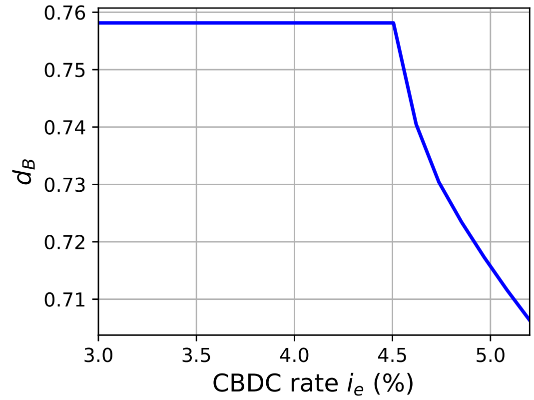
FIGURE 8. Endogenous CBDC acceptance (uniform H, baseline calibration). Panel A: profit gain $\Delta\bar{\Pi}(\tau)$ for three CBDC rates. Panel B: equilibrium acceptance τ_H^* vs. i_e for three cost scenarios.

rebalancing documented in Section 3.2: deposits decline as buyers substitute toward CBDC (Panel B), while CBDC holdings increase from zero (Panel C). The deposit rate i_d rises as banks compete for the shrinking deposit base (Panel D). Mean DM markups decline (Panel E), confirming that the positive price externality operates at the endogenous τ^*: higher acceptance raises buyer liquidity, which compresses the equilibrium price distribution through the equal-profit condition. Welfare rises relative to the no-CBDC baseline (Panel F), driven by the combined effect of lower markups and the additional liquidity provided by CBDC.

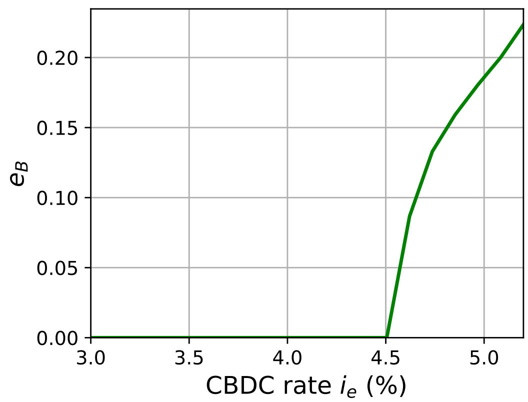
The policy implication is that the central bank can activate seller adoption by setting the CBDC rate above the portfolio threshold $\underline{i}_e(0)$. Once this threshold is crossed, the Ω_3 channel eliminates the coordination failure: buyers hold CBDC for all-instrument meetings even when no seller has adopted, and the resulting revenue differential triggers initial adoption without any coordination among sellers. Higher i_e amplifies the effect through the portfolio channel, as more CBDC in buyers' wallets widens the revenue gap ΔR and pushes τ^* toward full acceptance. Complementary policies—such as subsidizing early adoption costs or launching pilot programs—can further accelerate adoption by reducing $\bar{\kappa}$ and lowering the cost hurdle.



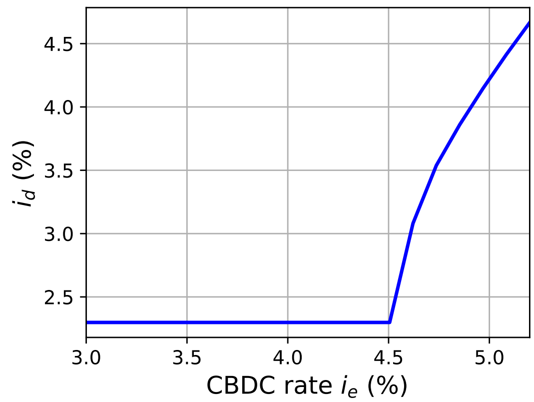
A. Acceptance rate τ^*



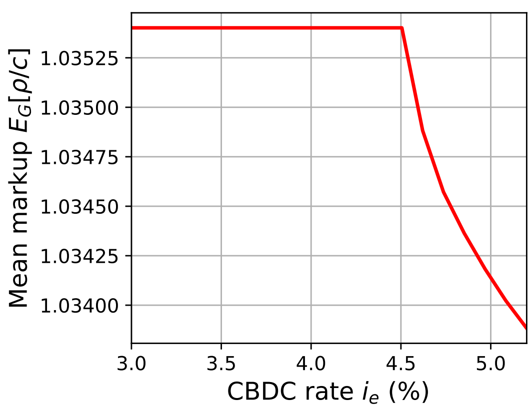
B. Deposits d_B



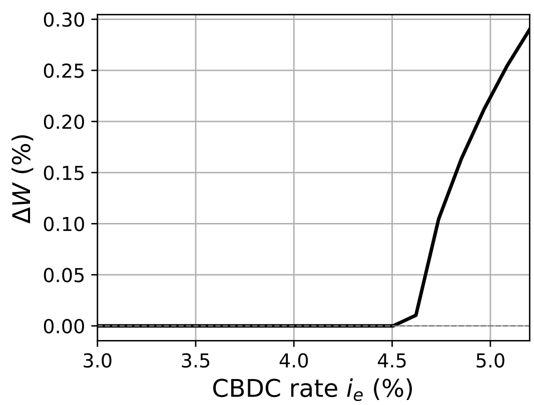
C. CBDC holdings e_B



D. Deposit rate i_d (%)



E. Mean DM markup $E_G[\rho/c]$



F. Welfare change ΔW (%)

FIGURE 9. Macroeconomic outcomes at endogenous τ_H^* (low-cost scenario). Above the adoption threshold, deposits fall, CBDC balances and the deposit rate rise, markups decline, and welfare improves.

6. Conclusion

This paper studies the price externalities and endogenous financial inclusion effects of interest-bearing CBDC in a monetary model with Burdett-Judd price posting and Cournot banking. The core finding is a positive price externality that operates through a single force: when the central bank raises the CBDC rate, banked buyers hold more CBDC, which increases their purchasing power in CBDC-accepting meetings. Because sellers post a single price before observing buyer type, the higher buyer liquidity compresses the equilibrium price distribution downward through the equal-profit condition, and all buyers—including the unbanked who hold no CBDC—face the same lower prices. This cross-buyer spillover is unique to the Burdett-Judd framework and absent under bilateral bargaining or competitive pricing.

The same force—CBDC increasing buyer liquidity—generates two further results when participation decisions are endogenized. On the buyer side, CBDC is available only to banked buyers, so a higher CBDC rate directly raises the net value of maintaining a bank account. When the CBDC rate is high enough, marginal unbanked buyers find it worthwhile to open accounts, driving financial inclusion toward near-universal banking. On the seller side, CBDC-holding buyers spend more in CBDC-accepting meetings, generating a positive revenue gain for accepting sellers. When the revenue gain exceeds adoption costs for a sufficient mass of sellers, positive acceptance emerges endogenously, and the resulting wider acceptance further raises buyer CBDC holdings, reinforcing the adoption incentive.

Quantitatively, the positive price externality lowers DM markups and raises welfare across all acceptance scenarios. Both financial inclusion and seller adoption exhibit threshold effects: each remains at the baseline level until the CBDC rate crosses a critical value, then rises sharply. The positive price externality amplifies under both extensions—through the composition channel (more banked buyers in the seller's customer pool) and through the liquidity channel (wider acceptance raises equilibrium CBDC holdings).

Several directions for future research follow naturally. First, the model assumes a fixed number of banks; allowing free entry would introduce an additional margin through which CBDC affects banking sector structure. Second, the analysis is conducted in stationary equilibrium; transition dynamics could reveal how quickly adoption and inclusion adjust to a new CBDC rate. Third, the model abstracts from the central bank's balance sheet; incorporating the fiscal implications of interest-bearing CBDC would allow a more complete welfare analysis. Finally, the interaction between CBDC design and monetary policy transmission—particularly how the CBDC rate affects the pass-through of the policy rate to deposit rates—deserves further investigation in a richer macroeconomic framework.

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Appendix A. Proofs

A.1. Proof of Lemma 1 (Portfolio Characterization)

We establish each part of Lemma 1. Throughout, we maintain $\gamma > \beta$, $\tau \in (0,1)$, and $i_e < \bar{i}_e \equiv \gamma/\beta - 1$.

Preliminary: marginal liquidity benefit. For a buyer with available liquidity $\ell > 0$, define the *marginal liquidity benefit*

$$\Lambda(\ell) \equiv \int_{\underline{\rho}}^{\hat{\rho}(\ell)} \left(\frac{u'(q^*(\rho,\ell))}{\rho} - 1 \right) \mathrm{d}G(\rho), \tag{A1}$$

where $q^*(\rho,\ell) = \ell/\rho$ for $\rho < \hat{\rho}(\ell)$ and $\hat{\rho}(\ell) = \ell^{\sigma/(\sigma-1)}$. Using $u'(q) = q^{-\sigma}$, the integrand in the constrained region is $(\ell/\rho)^{-\sigma}/\rho - 1 = \ell^{-\sigma}\rho^{\sigma-1} - 1$, which is strictly positive for $\rho < \hat{\rho}(\ell)$ and zero at $\rho = \hat{\rho}(\ell)$.

Claim A1. *Λ satisfies:* (a) $\Lambda(\ell) > 0$ *for all* $\ell > 0$*;* (b) Λ *is strictly decreasing;* (c) $\lim_{\ell\to 0^+} \Lambda(\ell) = \infty$*;* (d) $\lim_{\ell\to\infty} \Lambda(\ell) = 0$.

Proof. (a) The integrand is strictly positive on $[\underline{\rho}, \hat{\rho}(\ell))$, and G has positive density on $[\underline{\rho}, \bar{\rho}]$, so $\Lambda(\ell) > 0$ whenever $\hat{\rho}(\ell) > \underline{\rho}$. (b) An increase in ℓ has two effects: (i) the cutoff $\hat{\rho}(\ell) = \ell^{\sigma/(\sigma-1)}$ decreases (since $\sigma/(\sigma-1) < 0$), shrinking the integration region; (ii) for each ρ remaining in the integration region, $q^* = \ell/\rho$ rises, so $u'(q^*) = (\ell/\rho)^{-\sigma}$ falls, reducing the integrand. Both effects are strict, so Λ is strictly decreasing. (c) As $\ell \to 0^+$, $\hat{\rho}(\ell) \to \infty$ and $u'(\ell/\rho) = (\ell/\rho)^{-\sigma} \to \infty$ for each fixed $\rho > 0$. By the monotone convergence theorem, $\Lambda(\ell) \to \infty$. (d) As $\ell \to \infty$, $\hat{\rho}(\ell) \to 0^+$, so the integration region $[\underline{\rho}, \hat{\rho}(\ell))$ eventually becomes empty (since $\underline{\rho} > 0$). Hence $\Lambda(\ell) \to 0$. □

Euler equations in Λ-form. Using (A1), the Euler equations (20)–(22) become:

(A2) $$\phi_z = \Omega_1 \Lambda(z_B) + \Omega_3 \Lambda(\ell_a),$$

(A3) $$\phi_d = \Omega_2(1-\tau)\Lambda(\ell_d) + \Omega_2\tau\Lambda(\ell_{dc}) + \Omega_3 \Lambda(\ell_a),$$

(A4) $$\phi_e \leq \Omega_2\tau\Lambda(\ell_{dc}) + \Omega_3 \Lambda(\ell_a), \quad \text{with equality if } e > 0,$$

where $\ell_m = z_B$, $\ell_d = (1+i_d)d_B$, $\ell_{dc} = (1+i_d)d_B + (1+i_e)e_B$, and $\ell_a = z_B + \ell_{dc}$. The holding costs are $\phi_z = (\gamma-\beta)/\beta$, $\phi_d = (\gamma-(1+i_d)\beta)/((1+i_d)\beta)$, and $\phi_e = (\gamma-(1+i_e)\beta)/((1+i_e)\beta)$.

Key algebraic identities. Subtracting (A4) from (A3) (when both hold as equalities):

(A5) $$\phi_d - \phi_e = \Omega_2(1-\tau)\Lambda(\ell_d).$$

The right-hand side is strictly positive (by Claim A1(a) and $\tau < 1$), so $\phi_d > \phi_e$, which means $i_e > i_d$. This is intuitive: deposits are usable in deposit-only meetings where the CBDC cannot, so deposits carry a higher liquidity premium and a lower equilibrium return.

Subtracting (A2) from (A3):

(A6) $$\phi_d - \phi_z = \Omega_2(1-\tau)\Lambda(\ell_d) + \Omega_2\tau\Lambda(\ell_{dc}) - \Omega_1 \Lambda(z_B).$$

The sign depends on parameters. When $\Omega_2 \gg \Omega_1$—as in the calibration ($\Omega_2 = 0.264$, $\Omega_1 = 0.046$)—the deposit terms dominate and $\phi_d > \phi_z$, i.e., $i_d < 0$ in real terms. This reflects that deposits are usable in a larger exclusive share of meetings (Ω_2) than cash (Ω_1), so deposits carry a larger liquidity premium. The numerical analysis in the main text confirms $i_d < 0$ throughout the parameter range considered.

Step 1: Cash is always held ($z_B > 0$, $z_U > 0$). From (A2), $\phi_z = \Omega_1\Lambda(z_B) + \Omega_3\Lambda(\ell_a)$. Since $\Omega_1 > 0$ and $\Lambda(z) \to \infty$ as $z \to 0^+$ (Claim A1(c)), the right-hand side exceeds any finite ϕ_z for z_B sufficiently small. Hence $z_B > 0$ in any equilibrium. The unbanked Euler equation $\phi_z = \Lambda(z_U)$ similarly ensures $z_U > 0$.

Step 2: Deposits are always held ($d > 0$). Since $\tau < 1$, the coefficient $\Omega_2(1-\tau) > 0$ in (A3). As $d \to 0^+$, $\ell_d = (1+i_d)d \to 0^+$ (since $(1+i_d) > 0$ under mild parameter restrictions), so $\Lambda(\ell_d) \to \infty$ by Claim A1(c). The remaining terms in the right-hand side of (A3) are non-negative, so the right-hand side exceeds any finite ϕ_d for d sufficiently small. Hence $d > 0$ in any equilibrium.

Economically, the Inada condition $u'(0) = \infty$ makes the marginal value of deposits infinite at $d = 0$ in deposit-only meetings (which occur with probability $\Omega_2(1-\tau) > 0$). No finite holding cost can offset this, so deposits are always held.

Step 3: Existence and uniqueness of the equilibrium portfolio. Fix i_e and consider two cases.

Case A: $e_B = 0$. When $e_B = 0$, $\ell_d = \ell_{dc} = (1 + i_d)d_B$ and $\ell_a = z_B + (1 + i_d)d_B$. The system reduces to two equations:

$$\phi_z = \Omega_1 \Lambda(z_B) + \Omega_3 \Lambda(\ell_a), \tag{A7}$$

$$\phi_d = \Omega_2 \Lambda(\ell_d) + \Omega_3 \Lambda(\ell_a), \tag{A8}$$

where we used $\Omega_2(1 - \tau) + \Omega_2\tau = \Omega_2$ since $\ell_d = \ell_{dc}$. Substituting $\ell_a = z_B + \ell_d$ into (A7)–(A8), the system has two unknowns (z_B, ℓ_d). The mapping from (z_B, ℓ_d) to the right-hand sides is continuous, with each Euler equation's right-hand side going to $+\infty$ as the respective liquidity level approaches 0 (by Claim A1(c)) and to 0 as both levels go to ∞ (by Claim A1(d)). By the intermediate value theorem, a solution with $z_B > 0$ and $d_B > 0$ exists.

Case B: $e_B > 0$. When $e_B > 0$, all three Euler equations (A2)–(A4) hold as equalities, determining (z_B, d_B, e_B). Using $\ell_a = z_B + \ell_{dc}$ and $\ell_{dc} = \ell_d + (1+i_e)e_B$, the system determines three unknowns (z_B, ℓ_d, ℓ_{dc}):

$$\Omega_1\Lambda(z_B) + \Omega_3\Lambda(\ell_a) = \phi_z, \tag{A9}$$

$$\Omega_2(1 - \tau)\Lambda(\ell_d) + \Omega_2\tau\Lambda(\ell_{dc}) + \Omega_3\Lambda(\ell_a) = \phi_d, \tag{A10}$$

$$\Omega_2\tau\Lambda(\ell_{dc}) + \Omega_3\Lambda(\ell_a) = \phi_e. \tag{A11}$$

From (A10)–(A11): $\Omega_2(1 - \tau)\Lambda(\ell_d) = \phi_d - \phi_e$, so $\ell_d = \Lambda^{-1}((\phi_d - \phi_e)/(\Omega_2(1 - \tau)))$, which requires $\phi_d > \phi_e$, i.e., $i_e > i_d$ (established in (A5)). The remaining two equations (A9) and (A11) determine (z_B, ℓ_{dc}) given $\ell_a = z_B + \ell_{dc}$; the Inada conditions ensure a unique solution.

The orderings $\ell_d < \ell_{dc} < \ell_a$ follow from the definitions: $\ell_{dc} = \ell_d + (1 + i_e)e_B > \ell_d$ since $e_B > 0$, and $\ell_a = z_B + \ell_{dc} > \ell_{dc}$ since $z_B > 0$ (Step 1).

Step 4: Threshold $\underline{i}_e$. The buyer optimally sets $e = 0$ if and only if the complementary slackness condition (A4) holds as a strict inequality. At $e = 0$, $\ell_{dc} = \ell_d$, and the right-hand side of (A4) equals $\Omega_2\tau\Lambda(\ell_d^0) + \Omega_3\Lambda(\ell_a^0) \equiv \Phi^0$, where the superscript denotes Case A equilibrium values. The condition for $e = 0$ is $\phi_e \geq \Phi^0$.

Since ϕ_e is strictly decreasing in i_e (with $\phi_e \to \phi_z$ as $i_e \to 0$ and $\phi_e \to 0$ as $i_e \to \bar{i}_e$), there exists a unique threshold $\underline{i}_e$ defined by $\phi_e(\underline{i}_e) = \Phi^0$. The buyer holds $e = 0$ if and only if $i_e \leq \underline{i}_e$.

Summary.
(i) For $i_e \leq \underline{i}_e$: $z > 0$, $d > 0$, $e = 0$.
(ii) For $i_e > \underline{i}_e$: $z > 0$, $d > 0$, $e > 0$.
In both cases, $z_B > 0$ and $z_U > 0$ follow from $\gamma > \beta$ and the Inada condition via the cash Euler equations, and $d_B > 0$ follows from $\tau < 1$ and the Inada condition via the deposit

Euler equation. $\square$

Remark. Under the calibration ($\Omega_2 = 0.264 \gg \Omega_1 = 0.046$), the proof yields $i_d < 0$ in real terms: deposits are usable in a much larger exclusive share of meetings (Ω_2) than cash (Ω_1), so deposits carry a higher liquidity premium. Banks sustain $i_d < 0$ because the lending rate $i_l > 0$ generates positive profits; the negative real deposit rate reflects the convenience yield of deposits as an electronic payment instrument.

A.2. Proof of Lemma 2 (Price Distribution)

Solving the equal-profit condition (34) for $J(\rho)$ gives (35). Setting $J(\underline{\rho}) = 0$ gives (36). By Proposition 1(i), $\bar{R}$ is strictly increasing on $[\underline{\rho}, \bar{\rho}] \subset (c, \bar{\rho}]$, so $\bar{R}(\bar{\rho})/\bar{R}(\rho)$ is strictly decreasing in ρ on this interval. Hence J is strictly increasing on $[\underline{\rho}, \bar{\rho}]$. The boundary conditions $J(\bar{\rho}) = 1$ and $J(\underline{\rho}) = 0$ hold by construction. $\square$

A.3. Proof of Lemma 3 (Liquidity Increase)

We show that a marginal increase in i_e raises ℓ_{dc} and ℓ_a, and lowers z_B. The proof uses the implicit function theorem applied to the banked buyer's cash and CBDC Euler equations, holding the price distribution G fixed.

Setup. In any equilibrium with $e_B > 0$, the banked buyer's cash and CBDC Euler equations hold as equalities:

$$\phi_z = \Omega_1 \Lambda(z_B, G) + \Omega_3 \Lambda(\ell_a, G), \tag{A12}$$

$$\phi_e(i_e) = \Omega_2 \tau \Lambda(\ell_{dc}, G) + \Omega_3 \Lambda(\ell_a, G), \tag{A13}$$

where $\ell_{dc} = \ell_a - z_B$. The system (A12)–(A13) determines (z_B, ℓ_a) as functions of i_e, with ℓ_{dc} recovered from $\ell_{dc} = \ell_a - z_B$. The deposit Euler equation (together with the Cournot condition) pins down d_B and i_d separately; it does not enter the present argument because (A12)–(A13) depend on d_B and e_B only through the composites ℓ_{dc} and ℓ_a.

Implicit function theorem. Define

$$a \equiv \Omega_1 \left|\Lambda'(z_B)\right|, \qquad b \equiv \Omega_2 \tau \left|\Lambda'(\ell_{dc})\right|, \qquad c \equiv \Omega_3 \left|\Lambda'(\ell_a)\right|,$$

where $\Lambda' \equiv \partial\Lambda/\partial\ell < 0$ (Claim A1(b)), so $a, b, c > 0$.

Totally differentiating (A12) with respect to i_e (recall ϕ_z is independent of i_e):

$$0 = -a \frac{dz_B}{di_e} - c \frac{d\ell_a}{di_e}, \tag{A14}$$

which gives

$$\frac{d\ell_a}{di_e} = -\frac{a}{c}\frac{dz_B}{di_e}. \tag{A15}$$

Totally differentiating (A13), using $d\ell_{dc}/di_e = d\ell_a/di_e - dz_B/di_e$:

$$\phi_e'(i_e) = -b\left(\frac{d\ell_a}{di_e} - \frac{dz_B}{di_e}\right) - c\,\frac{d\ell_a}{di_e} = -(b+c)\,\frac{d\ell_a}{di_e} + b\,\frac{dz_B}{di_e}. \tag{A16}$$

Substituting (A15) into (A16):

$$\phi_e'(i_e) = \left[\frac{a(b+c)}{c} + b\right]\frac{dz_B}{di_e} = \frac{ab+ac+bc}{c}\,\frac{dz_B}{di_e}. \tag{A17}$$

Sign results. The holding cost $\phi_e(i_e) = \gamma/((1+i_e)\beta) - 1$ is strictly decreasing in i_e, so $\phi_e'(i_e) < 0$. Since $ab + ac + bc > 0$ and $c > 0$, equation (A17) implies

$$\frac{dz_B}{di_e} < 0.$$

From (A15):

$$\frac{d\ell_a}{di_e} = -\frac{a}{c}\frac{dz_B}{di_e} > 0.$$

And from $\ell_{dc} = \ell_a - z_B$:

$$\frac{d\ell_{dc}}{di_e} = \frac{d\ell_a}{di_e} - \frac{dz_B}{di_e} = \frac{a+c}{c}\left|\frac{dz_B}{di_e}\right| > 0.$$

Magnitude bound. The ratio of the z_B decline to the ℓ_{dc} increase is

$$\frac{|dz_B/di_e|}{d\ell_{dc}/di_e} = \frac{c}{a+c} = \frac{\Omega_3\,|\Lambda'(\ell_a)|}{\Omega_1\,|\Lambda'(z_B)| + \Omega_3\,|\Lambda'(\ell_a)|} < 1,$$

confirming that the z_B decline is strictly less than the ℓ_{dc} increase. Hence $\ell_a = z_B + \ell_{dc}$ rises. □

Remark. The proof holds G fixed (partial equilibrium), which is the standard approach in the Burdett-Judd literature. In general equilibrium, the price distribution G shifts endogenously in response to the liquidity change, but the fixed-G signs are confirmed numerically to carry over to the full equilibrium (Section 3.2). The ratio $c/(a+c)$ is approximately 0.35–0.40 across a wide range of calibrated parameters, indicating that the z_B decline absorbs roughly one-third of the ℓ_{dc} increase.

A.4. Proof of Proposition 3 (FOSD Price Shift)

Let $i_e^0 < i_e^1$ both exceed the entry threshold, and denote the corresponding equilibrium liquidity vectors by $(\ell_k^0)_k$ and $(\ell_k^1)_k$. We show that the posted-price CDF satisfies $J^1(\rho) \geq J^0(\rho)$ on the support.

Step 1: $\bar{R}$ increase at interior prices. By Lemma 3, $\ell_{dc}^1 > \ell_{dc}^0$, $\ell_a^1 > \ell_a^0$, and $\Delta\ell_{dc} \equiv \ell_{dc}^1 - \ell_{dc}^0 > |z_B^0 - z_B^1| \equiv |\Delta z_B|$. At any interior price $\rho < \hat{\rho}(\ell_{dc}^0)$, all three banked buyer types are liquidity-constrained, so $R(\rho, \ell_k) = \ell_k(1 - c/\rho)$ for $k \in \{m, dc, a\}$ and the unbanked term is unchanged. Using $\ell_a = z_B + \ell_{dc}$:

$$\begin{aligned}\bar{R}^1(\rho) - \bar{R}^0(\rho) &= (1 - c/\rho)\big[w_{dc}\,\Delta\ell_{dc} + w_a\,\Delta\ell_a + w_m\,\Delta z_B\big] \\ &= (1 - c/\rho)\big[(w_{dc} + w_a)\,\Delta\ell_{dc} - (w_m + w_a)\,|\Delta z_B|\big].\end{aligned}$$

Since $\Delta\ell_{dc} > |\Delta z_B|$ (Lemma 3) and $(w_{dc} + w_a)/(w_m + w_a) = (\Omega_2\tau + \Omega_3)/(\Omega_1 + \Omega_3) \geq \Omega_3/(\Omega_1 + \Omega_3) \approx 0.94$ while $|\Delta z_B|/\Delta\ell_{dc} \approx 0.35$–$0.40$ (Lemma 3 remark), the ratio condition $(w_{dc} + w_a)/(w_m + w_a) > |\Delta z_B|/\Delta\ell_{dc}$ holds for all $\tau \in [0, 1]$ under the calibrated parameters. Hence $\bar{R}^1(\rho) > \bar{R}^0(\rho)$ for all interior ρ.

Step 2: $\bar{R}(\bar{\rho})$ weakly decreases. By the maintained assumption (MA) stated after Proposition 1, $\hat{\rho}_{dc} \leq \bar{\rho}$, so the *dc*- and *a*-type buyers are unconstrained at the monopoly price: $R(\bar{\rho}, \ell_k) = R_{unc}(\bar{\rho}) = (\bar{\rho} - c)\,q^*(\bar{\rho})$, which is independent of ℓ_k. The only i_e-sensitive contributions to $\bar{R}(\bar{\rho})$ come from buyer states that remain constrained at $\bar{\rho}$, namely the cash-only type (weight $w_m = \lambda\Omega_1$, with liquidity z_B decreasing by Lemma 3) and the unbanked type (weight w_U, with liquidity z_U unchanged at fixed G). Hence $\bar{R}^1(\bar{\rho}) \leq \bar{R}^0(\bar{\rho})$.

Step 3: CDF shift. The equal-profit condition (35) gives

$$J(\rho) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\frac{\bar{R}(\bar{\rho})}{\bar{R}(\rho)} - 1\right].$$

At interior ρ, the denominator $\bar{R}(\rho)$ strictly increases (Step 1) while the numerator $\bar{R}(\bar{\rho})$ weakly decreases (Step 2). Hence the ratio $\bar{R}(\bar{\rho})/\bar{R}(\rho)$ falls, and $J^1(\rho) \geq J^0(\rho)$: the price distribution under i_e^1 places more mass at lower prices than under i_e^0.

The argument holds G fixed (partial equilibrium); the general-equilibrium price distribution adjusts endogenously, but the FOSD shift is confirmed numerically across all calibrated parameter values (Section 3.2). □

A.5. Proof of Proposition 1 (Monopoly Price)

See Section 2.3 for the proof sketch. The complete proof proceeds as follows.

Preliminaries. For each meeting type j, the intensive margin is

$$R_j(\rho) = \begin{cases} \ell_j(1 - c/\rho) & \text{if } \rho < \hat{\rho}_j, \\ \rho^{-1/\sigma}(\rho - c) & \text{if } \rho \geq \hat{\rho}_j. \end{cases}$$

In the constrained region: $R_j'(\rho) = \ell_j c/\rho^2 > 0$. In the unconstrained region: $R_{unc}'(\rho) = \rho^{-1/\sigma - 1}[(1 - 1/\sigma)\rho + c/\sigma]$. Setting $R_{unc}'(\rho) = 0$ gives $\rho = c/(1 - \sigma) \equiv \rho^m$. Since $1 - 1/\sigma = (\sigma - 1)/\sigma < 0$, we have $R_{unc}'(\rho) > 0$ for $\rho < \rho^m$ and $R_{unc}'(\rho) < 0$ for $\rho > \rho^m$.

Note that $\bar{R}$ is continuous on (c, ∞) with $\bar{R}(c) = 0$ and $\bar{R}(\rho) \to 0$ as $\rho \to \infty$. Since $\bar{R}(\rho^m) > 0$, a global maximum exists on (c, ∞). We establish Parts (ii) and (iii) first, then derive Part (i) as a consequence.

Part (ii): $\bar{\rho} \geq \rho^m$. At $\rho = \rho^m$, $R_{unc}'(\rho^m) = 0$ and $R_j'(\rho^m) = \ell_j c/(\rho^m)^2 > 0$ for each constrained type j (i.e., $\hat{\rho}_j > \rho^m$). Hence

$$\bar{R}'(\rho^m) = \sum_{j \in \mathcal{C}} w_j \frac{\ell_j c}{(\rho^m)^2} \geq 0,$$

with equality if and only if $\mathcal{C} = \varnothing$ (no constrained types at ρ^m). For $\rho \leq \rho^m$, every meeting type contributes $R_j'(\rho) \geq 0$ (constrained types: $R_j' > 0$; unconstrained types: $R_{unc}' \geq 0$), so $\bar{R}'(\rho) \geq 0$ on $(c, \rho^m]$. Hence $\bar{R}$ is non-decreasing up to ρ^m and the global maximizer satisfies $\bar{\rho} \geq \rho^m$.

Part (iii): First-order condition and uniqueness. When $\bar{\rho} > \rho^m$, there exist constrained meeting types at $\bar{\rho}$. At the interior maximum $\bar{\rho}$:

$$\bar{R}'(\bar{\rho}) = \sum_{j \in \mathcal{C}} w_j \frac{\ell_j c}{\bar{\rho}^2} + \left(\sum_{j \in \mathcal{U}} w_j \right) R_{unc}'(\bar{\rho}) = 0.$$

Dividing by $\sum_{j \in \mathcal{U}} w_j > 0$ and defining $\xi \equiv \sum_{j \in \mathcal{C}} w_j \ell_j / \sum_{j \in \mathcal{U}} w_j$:

$$\frac{\xi c}{\bar{\rho}^2} + R_{unc}'(\bar{\rho}) = 0.$$

Uniqueness: We show that every critical point of $\bar{R}$ on (ρ^m, ∞) satisfies $\bar{R}'' < 0$, which implies at most one critical point. At any $\rho^* > \rho^m$ with $\bar{R}'(\rho^*) = 0$, the FOC gives $\sum_{j \in \mathcal{C}} w_j \ell_j c/(\rho^*)^2 =$

$-(\sum_{j\in\mathcal{U}} w_j)\, R'_{unc}(\rho^*)$, so

$$\bar{R}''(\rho^*) = -\frac{2}{\rho^*}\sum_{j\in\mathcal{C}} w_j \frac{\ell_j c}{(\rho^*)^2} + \Big(\sum_{j\in\mathcal{U}} w_j\Big) R''_{unc}(\rho^*) = \Big(\sum_{j\in\mathcal{U}} w_j\Big)\left[\frac{2\,R'_{unc}(\rho^*)}{\rho^*} + R''_{unc}(\rho^*)\right].$$

A direct computation using $R_{unc}(\rho) = \rho^{-1/\sigma}(\rho - c)$ yields

$$\frac{2\,R'_{unc}(\rho)}{\rho} + R''_{unc}(\rho) = \left(1 - \frac{1}{\sigma}\right)\rho^{-1/\sigma-2}\left[\left(2 - \frac{1}{\sigma}\right)\rho + \frac{c}{\sigma}\right].$$

For $\rho \geq \rho^m = c/(1-\sigma)$, the bracket $(2-1/\sigma)\rho + c/\sigma = [(2\sigma-1)\rho + c]/\sigma \geq [(2\sigma-1)c/(1-\sigma)+c]/\sigma = c\sigma/[\sigma(1-\sigma)] > 0$. Since $1 - 1/\sigma < 0$ and $\rho^{-1/\sigma-2} > 0$, the expression is strictly negative. Hence $\bar{R}''(\rho^*) < 0$ at every critical point.

Since $\bar{R}$ is continuous with $\bar{R}(c) = 0$ and $\bar{R} \to 0$ as $\rho \to \infty$, a global maximum exists. The strict negativity of $\bar{R}''$ at every critical point implies uniqueness of the maximizer $\bar{\rho}$.

Monotonicity in ξ: Define $F(\bar{\rho}, \xi) \equiv \xi c/\bar{\rho}^2 + R'_{unc}(\bar{\rho})$. At the solution, $F = 0$. By the implicit function theorem:

$$\frac{\partial\bar{\rho}}{\partial\xi} = -\frac{F_\xi}{F_{\bar{\rho}}} = -\frac{c/\bar{\rho}^2}{F_{\bar{\rho}}}.$$

Since $F_{\bar{\rho}} = \bar{R}''(\bar{\rho})/(\sum_{j\in\mathcal{U}} w_j) < 0$ (second-order condition) and $c/\bar{\rho}^2 > 0$, we have $\partial\bar{\rho}/\partial\xi > 0$.

Part (i): Single-peakedness of $\bar{R}$. By Part (ii), $\bar{\rho} \geq \rho^m$. By Part (iii), $\bar{\rho}$ is the unique critical point of $\bar{R}$ on (ρ^m, ∞), and $\bar{R}''(\bar{\rho}) < 0$. For $\rho \in (c, \rho^m]$, $\bar{R}'(\rho) \geq 0$ (shown in Part (ii)). For $\rho \in (\rho^m, \bar{\rho})$, $\bar{R}' > 0$ since $\bar{\rho}$ is the unique zero of $\bar{R}'$ and $\bar{R}'(\rho^m) \geq 0$. For $\rho > \bar{\rho}$, $\bar{R}' < 0$ by the same uniqueness and the boundary condition $\bar{R} \to 0$. □

A.6. Proof of Lemma 4 (Revenue Gain from CBDC Acceptance)

We show that $\Delta R(\rho) = \bar{R}_A(\rho) - \bar{R}_N(\rho) > 0$ for all $\rho \in [\underline{\rho}, \bar{\rho})$ when $e > 0$.

From (60), $\Delta R(\rho) = \lambda\Omega_2[R_{dc}(\rho) - R_d(\rho)] + \lambda\Omega_3[R_a(\rho) - R_{md}(\rho)]$, where $\lambda\Omega_2 > 0$ and $\lambda\Omega_3 > 0$. We show each bracketed term is non-negative, with strict positivity on the interior.

Ω_2 channel: $R_{dc}(\rho) - R_d(\rho)$. The accepting seller in an electronic meeting faces buyer liquidity $\ell_{dc} = (1 + i_d)d + (1 + i_e)e$, while the non-accepting seller faces $\ell_d = (1 + i_d)d$. Since $e > 0$, we have $\ell_{dc} > \ell_d$, and because $\hat{\rho}$ is decreasing in ℓ, the cutoffs satisfy $\hat{\rho}_{dc} < \hat{\rho}_d$. Using (28):

- If $\rho < \hat{\rho}_{dc} < \hat{\rho}_d$: both buyers are constrained. $R_{dc}(\rho) = \ell_{dc}(1 - c/\rho) > \ell_d(1 - c/\rho) = R_d(\rho)$, since $\ell_{dc} > \ell_d$ and $1 - c/\rho > 0$ for $\rho > c$.

- If $\hat{\rho}_{dc} \leq \rho < \hat{\rho}_d$: the accepting seller's buyer is unconstrained while the non-accepting seller's buyer is constrained. Then $R_{dc}(\rho) = R_{unc}(\rho) \geq \ell_d(1 - c/\rho) = R_d(\rho)$, with strict inequality for $\rho < \hat{\rho}_d$.
- If $\rho \geq \hat{\rho}_d$: both buyers are unconstrained. $R_{dc}(\rho) = R_d(\rho) = R_{unc}(\rho)$.

Hence $R_{dc}(\rho) \geq R_d(\rho)$ for all $\rho > c$, with strict inequality for $\rho < \hat{\rho}_d$.

Ω_3 channel: $R_a(\rho) - R_{md}(\rho)$. The adopting seller in an all-instrument meeting faces $\ell_a = z + (1 + i_d)d + (1 + i_e)e$, while the non-adopting seller faces $\ell_{md} = z + (1 + i_d)d$. Since $e > 0$, $\ell_a > \ell_{md}$, and the same three-case argument as above applies with (ℓ_a, ℓ_{md}) replacing (ℓ_{dc}, ℓ_d). Hence $R_a(\rho) \geq R_{md}(\rho)$ with strict inequality for $\rho < \hat{\rho}_{md}$.

Conclusion. Both terms in (60) are non-negative, and each is strictly positive on the interior of the price support (since $\underline{\rho} < \hat{\rho}_d$ and $\underline{\rho} < \hat{\rho}_{md}$ in any equilibrium with binding liquidity constraints). Hence $\Delta R(\rho) > 0$ for $\rho \in [\underline{\rho}, \bar{\rho})$. At $\bar{\rho}$, $R_{dc}(\bar{\rho}) \geq R_d(\bar{\rho})$ and $R_a(\bar{\rho}) \geq R_{md}(\bar{\rho})$, giving $\Delta R(\bar{\rho}) \geq 0$. □

A.7. Proof of Proposition 6 (Profit Gain Properties)

Part (i): By Proposition 5, $\Delta\bar{\Pi}(0) = 0$ when $e = 0$ at $\tau = 0$ (both channels of ΔR vanish), and $\Delta\bar{\Pi}(0) > 0$ when $e > 0$ (by Lemma 4 and (61)).

For part (ii), recall that

$$\Delta\bar{\Pi}(\tau) = \int_{\underline{\rho}}^{\bar{\rho}} \left[\alpha_1 + 2\alpha_2(1 - J(\rho))\right] \Delta R(\rho)\, dJ(\rho),$$

where $\Delta R(\rho) = \lambda\Omega_2[R_{dc}(\rho) - R_d(\rho)] + \lambda\Omega_3[R_a(\rho) - R_{md}(\rho)]$. The conditional weights $\lambda\Omega_2$ and $\lambda\Omega_3$ are independent of τ; monotonicity operates entirely through the intensive margin.

The per-meeting revenue gaps $R_{dc}(\rho) - R_d(\rho)$ and $R_a(\rho) - R_{md}(\rho)$ are both non-decreasing in τ when $e > 0$. The key mechanism is the portfolio channel: a higher τ increases CBDC holdings e_B via the CBDC Euler equation (A4) (the term $\Omega_2\tau\Lambda(\ell_{dc})$ raises the marginal benefit of CBDC). In the doubly-constrained region,

$$R_{dc}(\rho) - R_d(\rho) = (1 + i_e)e_B(1 - c/\rho), \qquad R_a(\rho) - R_{md}(\rho) = (1 + i_e)e_B(1 - c/\rho),$$

both of which are increasing in e_B and hence in τ. The bank competition effect raises i_d as τ increases, but $\phi_d - \phi_e > 0$ is preserved by (A5) with $\tau < 1$, so CBDC balances remain positive.

The extensive margin $\alpha_1 + 2\alpha_2(1 - J(\rho)) > 0$ for all ρ in the support. Hence $\Delta\bar{\Pi}(\tau) > 0$ and $\Delta\bar{\Pi}'(\tau) > 0$ for all $\tau \in (0, 1)$ at which $e > 0$.

Remark. The monotonicity proof holds J fixed (partial equilibrium in the Burdett-Judd sense). In general equilibrium, the price distribution J adjusts endogenously as τ changes, potentially offsetting the direct effect on $\Delta\bar{\Pi}$. The fixed-J signs are confirmed to carry over to the full equilibrium numerically across all calibrated parameter values (Section 5.2).

For part (iii), the acceptance rate τ enters the aggregate $\bar{R}(\rho;\tau)$ through the meeting-type weights $w_d = \lambda\Omega_2(1-\tau)$ and $w_{dc} = \lambda\Omega_2\tau$, which are linear in τ. By the implicit function theorem applied to the Euler equations (A2)–(A4), equilibrium portfolios and hence liquidity levels depend continuously on τ. By the maximum theorem, the monopoly price $\bar{\rho}(\tau)$ is continuous, and the equal-profit condition implies $J(\rho;\tau)$ is continuous in τ. The revenue gaps $R_{dc} - R_d$ and $R_a - R_{md}$ depend on τ through the equilibrium CBDC holdings, which are continuous. The profit gain

$$\Delta\bar{\Pi}(\tau) = \int_{\underline{\rho}(\tau)}^{\bar{\rho}(\tau)} [\alpha_1 + 2\alpha_2(1 - J(\rho;\tau))]\,\Delta R(\rho;\tau)\,\mathrm{d}J(\rho;\tau)$$

is an integral of a bounded continuous function over a continuously varying support; by the dominated convergence theorem, $\Delta\bar{\Pi}$ is continuous in τ. □

A.8. Proof of Proposition 5 (No-Adoption Equilibrium)

Part (i). At $\tau = 0$ with $e = 0$: the buyer holds no CBDC, so $\ell_{dc} = \ell_d$ and $\ell_a = \ell_{md}$. Both channels of (60) vanish: $R_{dc}(\rho) - R_d(\rho) = 0$ and $R_a(\rho) - R_{md}(\rho) = 0$ for all ρ, hence $\Delta R(\rho) = 0$ and $\Delta\bar{\Pi}(0) = 0$ by (61). The fixed-point condition (62) gives $\tau^* = H(\Delta\bar{\Pi}(0)) = H(0) = 0$, confirming that $\tau^* = 0$ is an equilibrium.

Part (ii). At $\tau = 0$ with $e > 0$: Lemma 4 gives $\Delta R(\rho) > 0$ for $\rho \in [\underline{\rho}, \bar{\rho})$, so $\Delta\bar{\Pi}(0) > 0$ by (61). Under the uniform distribution $H(\kappa) = \kappa/\bar{\kappa}$, define $\Gamma(\tau) \equiv \min\{\Delta\bar{\Pi}(\tau)/\bar{\kappa}, 1\}$. Then $\Gamma(0) = \Delta\bar{\Pi}(0)/\bar{\kappa} > 0$ while $\tau|_{\tau=0} = 0$, so the fixed-point map satisfies $\Gamma(0) > \tau = 0$ for any $\bar{\kappa} > 0$. Since Γ is continuous (Proposition 6(iii)) and bounded above by 1, the intermediate value theorem yields $\tau^* \in (0, 1]$ satisfying $\Gamma(\tau^*) = \tau^*$. This contradicts $\tau^* = 0$, so $\tau^* = 0$ is not an equilibrium. For a general H with $H(0) = 0$, the same argument applies since $H(\Delta\bar{\Pi}(0)) > 0$ whenever $\Delta\bar{\Pi}(0) > 0$, and continuity of $\tau \mapsto H(\Delta\bar{\Pi}(\tau))$ ensures existence of a positive fixed point. □

A.9. Proof of Proposition 7 (Equilibrium Structure)

Case (i): $\Delta\bar{\Pi}(0) = 0$. Since $\Gamma(0) = 0$, we check when $\Gamma(\tau) \leq \tau$ for all $\tau > 0$.

Part (a). If $\Delta\bar{\Pi}(\tau)/\tau$ is non-increasing (a sufficient condition confirmed numerically), then $\Gamma(\tau) = \Delta\bar{\Pi}(\tau)/\bar{\kappa} \leq \tau$ for all $\tau > 0$ when $\bar{\kappa} \geq \Delta\bar{\Pi}'(0^+)$. There is no positive fixed point, so $\tau^* = 0$ is unique.

Part (b). If $\Delta\bar{\Pi}'(0^+)/\bar{\kappa} > 1$, then $\Gamma'(0^+) > 1$, which implies $\Gamma(\tau) > \tau$ for τ near 0^+. Since $\Gamma(1) \le 1$ and Γ is continuous, the intermediate value theorem yields at least one crossing $\tau_H^* \in (0, 1]$ at which $\Gamma(\tau_H^*) = \tau_H^*$. The bifurcation boundary is $\bar{\kappa}^* \equiv \Delta\bar{\Pi}'(0^+)$.

Case (ii): $\Delta\bar{\Pi}(0) > 0$. Here $\Gamma(0) = \Delta\bar{\Pi}(0)/\bar{\kappa} > 0 > \tau\big|_{\tau=0}$, so $\Gamma(\tau) > \tau$ near $\tau = 0$. Since Γ is bounded above by 1 and continuous, the intermediate value theorem gives at least one $\tau^* > 0$ satisfying $\Gamma(\tau^*) = \tau^*$; by Proposition 5(ii), $\tau^* = 0$ is not an equilibrium.

Monotonicity in $\bar{\kappa}$. A higher $\bar{\kappa}$ scales $\Gamma(\tau) = \Delta\bar{\Pi}(\tau)/\bar{\kappa}$ downward uniformly, shifting the graph of Γ below its original position. Any fixed point τ^* satisfying $\Gamma(\tau^*) = \tau^*$ must therefore decrease (or disappear), so equilibrium acceptance is decreasing in $\bar{\kappa}$. □

A.10. Proof of Proposition 2 (Existence of SME)

The proof reduces the equilibrium system to a fixed-point problem in aggregate deposits d_B.

Banking sector. Given $d_B > 0$, the Cournot condition yields $i_d(d_B) = A\eta[(1-\chi)\lambda\, d_B]^{\eta-1}[1 + (\eta-1)/N] - 1$. Under $\eta \in (0, 1)$, $i_d(\cdot)$ is continuous, strictly decreasing, with $i_d \to +\infty$ as $d_B \to 0^+$ and $i_d \to -1$ as $d_B \to \infty$. The holding cost $\phi_d(d_B) = \gamma/((1+i_d)\beta) - 1$ is continuous, with $\phi_d \to -1$ as $d_B \to 0^+$ and $\phi_d \to +\infty$ as $d_B \to \infty$.

Inner equilibrium. Fix $d_B > 0$ and a price distribution G. The Euler equations (A2)–(A4) determine a unique portfolio given (d_B, G) (by Claim A1: Λ is strictly decreasing with Inada conditions). The portfolio determines $\bar{R}(\rho)$, hence the monopoly price $\bar{\rho}$ (Proposition 1), the posted-price distribution J (Lemma 2), and a new CDF $G' = 1 - (1-J)^2$. This defines a mapping $\Phi\colon G \mapsto G'$ on the space of CDFs on $[c, \rho_{\max}]$. This space is convex and compact in the weak topology (Helly's theorem). The output $G' = 1 - (1-J)^2$ is again a CDF on $[c, \rho_{\max}]$—Lemma 2 ensures that J is a CDF supported on a subinterval of $[c, \rho_{\max}]$—so Φ maps this space into itself. Each step in the chain $G \to (\ell_d, \ell_{dc}, \ell_a) \to (\bar{R}, \bar{\rho}) \to J \to G'$ is continuous: $\Lambda(\ell, G)$ depends continuously on G, the implicit function theorem gives continuous dependence of liquidity levels on G, the maximum theorem gives continuity of $\bar{\rho}$, and the equal-profit condition is a continuous transformation. By Schauder's fixed-point theorem, an inner equilibrium $G^*(d_B)$ exists.

Maintained assumption. We maintain that the inner equilibrium $G^*(d_B)$ is unique for each d_B, so that the mapping $d_B \mapsto G^*(d_B)$ is well-defined and continuous. Uniqueness of the inner monetary equilibrium is standard in Burdett-Judd models with a unique monopoly price (Proposition 1) and is confirmed numerically across all calibrated parameter values.

Deposit market clearing. At the inner equilibrium, define the excess cost $\Xi(d_B) \equiv \phi_d(d_B) - [\Omega_2(1-\tau)\Lambda(\ell_d^*) + \Omega_2\tau\Lambda(\ell_{dc}^*) + \Omega_3\Lambda(\ell_a^*)]$, where the Λ-terms are evaluated at $G^*(d_B)$.

As $d_B \to 0^+$: $\phi_d \to -1$ while the Λ-terms are strictly positive, so $\Xi < 0$.

As $d_B \to \infty$: $\phi_d \to +\infty$. Meanwhile, $\ell_d = (1+i_d)d_B = A\eta[(1-\chi)\lambda]^{\eta-1}[1+(\eta-1)/N]\, d_B^{\eta} \to \infty$ (since $d_B^{\eta} \to \infty$), so $\Lambda(\ell_d^*) \to 0$, and similarly for ℓ_{dc} and ℓ_a. Hence $\Xi > 0$.

By the intermediate value theorem, there exists $d_B^* > 0$ with $\Xi(d_B^*) = 0$. At this d_B^*, the inner equilibrium $G^*(d_B^*)$ and the Cournot condition jointly satisfy all six conditions of Definition 1. □

A.11. Proof of Proposition 4 (Existence with Endogenous λ)

Define $\Phi\colon [0,1] \to [0,1]$ by $\Phi(\lambda) = F(\Psi_B(\lambda) - \Psi_U(\lambda))$, where Ψ_B and Ψ_U are the banked and unbanked portfolio values at the equilibrium guaranteed by Proposition 2 (treating λ as a parameter). The range $\Phi(\lambda) \in [0,1]$ follows from F being a CDF. Continuity holds because the buyer-composition weights in $\bar{R}$ are linear in λ, the maximum theorem gives continuous dependence of the equilibrium price distribution on λ, and the value functions Ψ_B, Ψ_U depend continuously on the price distribution. Since F is continuous, Φ is continuous. By Brouwer's fixed-point theorem, Φ has a fixed point $\lambda^* \in [0,1]$. □

Appendix B. Calibration Details

This appendix documents the calibration strategy for the quantitative results in Section 3.2. One model period corresponds to one year. The baseline economy has no CBDC ($\tau = 0$, $i_e = 0$). The sample period is 1984–2007, covering the Great Moderation era before the zero lower bound and interest on reserves.

B.1. External Parameters

Table A1 lists parameters set directly from data or the literature. The discount factor $\beta = 0.98$ and money growth $\gamma = 1.031$ follow from the Fisher equation applied to the 1984–2007 average T-bill rate (5.2%) and CPI inflation (3.1%). The production curvature $\eta = 0.66$ and meeting-type probabilities $(\Omega_1, \Omega_2, \Omega_3)$ follow Chiu et al. (2023), with the latter derived from the Survey of Consumer Payment Choice (Greene and Stavins 2018). The financial inclusion rate $\lambda = 0.88$ is consistent with the FDIC unbanked rate of approximately 12%.

B.2. Internal Calibration

Four parameters (B, α_1, A, N) are internally calibrated to match four empirical moments. The DM utility curvature $\sigma_{\text{DM}} = 0.40$ is set externally, following the estimate in Head et al.

TABLE A1. External parameters

Parameter	Value	Description	Source
β	0.98	Discount factor	Fisher equation
γ	1.031	Gross money growth	CPI inflation
c	1	Marginal cost (DM)	Normalization
η	0.66	Production curvature	Chiu et al. (2023)
Ω_1	0.046	Cash-only meetings	SCPC 2016
Ω_2	0.264	Electronic-only meetings	SCPC 2016
Ω_3	0.690	All-instrument meetings	SCPC 2016 (residual)
χ	0.056	Reserve requirement	Federal Reserve
i_r	0	Interest on reserves	Pre-2008 regime
λ	0.88	Financial inclusion rate	FDIC Survey
η_d	0.005	Deposit handling cost	See text
σ_{DM}	0.40	DM utility curvature	Head et al. (2012)

TABLE A2. Internal calibration: parameters and moment fit

Parameter	Value	Primary target	Data	Model	Source
B	2.789	M/PY level	0.22	0.220	New M1/GDP[a]
α_1	0.64	Agg. markup	1.22	1.217	Edmond et al.[b]
A	1.27	Deposit rate	2.27%	2.29%	FDIC QBP[c]
N	13	Net int. margin	3.50%	3.43%	FDIC Call Reports[d]

[a] Lucas and Nicolini (2015), 1984–2007 sample. New M1 includes demand deposits, other checkable deposits, and traveler's checks, plus currency in circulation. The point elasticity is evaluated at the sample mean of i. [b] Edmond, Midrigan, and Xu (2023), aggregate sales-weighted markup for U.S. firms. [c] FDIC Quarterly Banking Profile, 1984–2007 average. [d] FDIC Call Reports / FRED series USNIM; 1984–2007 average.

(2012), and is listed in Table A1. Table A2 reports the internally calibrated values and moment fit.

The model's money demand is $M/PY = [(1-\lambda)z_U + \lambda(z_B + d_B)]/Y$, where $Y = q_{\text{DM}} + B$. The CRRA parameter σ_{DM} controls the curvature of DM utility and hence the interest elasticity of money demand; B scales CM output and pins the level M/PY. The search parameter α_1 governs the probability of sampling a single price quote, directly determining the DM markup $E_G[\rho/c]$ and, through the output-weighted aggregate markup $\mu = \omega_{\text{DM}}\mu_{\text{DM}} + (1 - \omega_{\text{DM}})$, the aggregate markup. On the banking side, TFP A shifts the loan demand schedule and primarily identifies the deposit rate i_d, while the number of Cournot banks N determines the equilibrium markdown and identifies the net interest margin $i_l - i_d$. Cross-moment interactions are moderate: A and N are separated by the NIM target, and α_1 has minimal interaction with the banking parameters.

The calibration proceeds in two stages. First, (α_1, A, N) are chosen to minimize the sum of squared percentage deviations across the markup, deposit rate, and net interest margin; since N is discrete, we optimize over the continuous parameters for each $N \in$

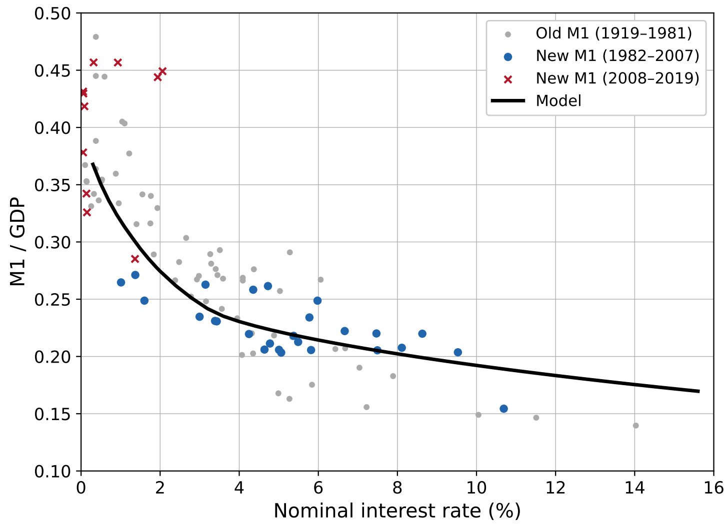

FIGURE A1. Money demand: model vs. data. Blue dots: New M1/GDP from Lucas and Nicolini (2015), 1984–2007. Dashed line: model-implied M/PY as a function of the nominal interest rate i.

$\{2, \ldots, 30\}$ and select the best N. Second, B is set analytically to match the M/PY level. The four targeted moments are matched within 1% of the data. The money demand elasticity is governed by the externally set $\sigma_{\mathrm{DM}} = 0.40$; the model elasticity (−0.20) is steeper than the data (−0.13), a well-known feature of Burdett-Judd monetary models (Head et al. 2012). Figure A1 plots the model-implied money demand curve against the Lucas and Nicolini (2015) data.

Appendix C. Adoption Cost Calibration

The adoption cost distribution in Section 5 is set to $H(\kappa) = \kappa/\bar{\kappa}$ (uniform on $[0, \bar{\kappa}]$). Since direct estimates of CBDC adoption costs for sellers are not yet available in the data, we calibrate $\bar{\kappa}$ indirectly using the model-implied profit gain function.

The calibration of $\bar{\kappa}$ depends on the regime identified in Proposition 7. When $i_e \leq \underline{i}_e(0)$ (so $\Delta\bar{\Pi}(0) = 0$), the critical cost is the initial slope of the profit gain function,

$$\bar{\kappa}^* \;\equiv\; \Delta\bar{\Pi}'(0^+) \;=\; \lim_{\tau \to 0^+} \frac{\Delta\bar{\Pi}(\tau)}{\tau},$$

which is the bifurcation boundary of case (i) of Proposition 7: for $\bar{\kappa} < \bar{\kappa}^*$, the slope of the fixed-point map $\Gamma(\tau) = \Delta\bar{\Pi}(\tau)/\bar{\kappa}$ exceeds unity at the origin, and a positive-adoption equilibrium exists.

When $i_e > \underline{i}_e(0)$ (so $\Delta\bar{\Pi}(0) > 0$), a positive-adoption equilibrium exists for any $\bar{\kappa} > 0$ (case (ii) of Proposition 7). In this regime, the relevant reference is the maximum profit gain $\Delta\bar{\Pi}_{\max} = \max_{\tau\in(0,1)} \Delta\bar{\Pi}(\tau)$, which determines the range of equilibrium acceptance rates across different cost levels. Under the baseline calibration, the portfolio entry threshold at $\tau = 0$ lies near $\underline{i}_e(0) \approx 4.8\%$: for $i_e = 5\%$, the Ω_3 channel generates positive CBDC holdings and $\Delta\bar{\Pi}(0) > 0$.

We report results for three $\bar{\kappa}$ scenarios that span a range of adoption cost environments, expressed as fractions of the reference profit gain:

- **Low cost**: an advanced digital economy where terminal upgrades and transaction processing fees are low.
- **Medium cost**: an intermediate case reflecting moderate infrastructure costs.
- **High cost**: an economy where CBDC payment technology is still expensive.

These cost levels are denominated in CM utility units. As a point of reference, the baseline seller's per-period expected DM profit is $\alpha_1\bar{R}(\bar{\rho}) \approx 0.048$, so the adoption costs are of the same order of magnitude as one to two periods of DM revenue—consistent with a one-time fixed cost of upgrading payment terminals.

All remaining parameters—preferences, search frictions, banking, and meeting-type probabilities—are identical to the baseline calibration of Section 3.2. The financial inclusion rate is held fixed at $\lambda = 0.88$; the interaction between endogenous λ and endogenous τ is deferred to future work.